
UNIFYING FORMAL EXPLANATIONS: A COMPLEXITY-THEORETIC PERSPECTIVE

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ABSTRACT

A substantial body of prior work has interrogated the computational complexity of deriving two foundational classes of explanations for the predictions of ML models: (i) *sufficient reasons*, namely subsets of input features whose fixation suffices to determine a prediction, and (ii) *contrastive reasons*, namely subsets of input features whose modification suffices to overturn a prediction. These explanatory notions have hitherto been scrutinized under a variety of disparate regimes — among them the non-probabilistic versus probabilistic dichotomy and the local versus global distinction. In the present study, we advance a unifying framework within which all such notions may be jointly analyzed, establishing that each is characterizable as the minimization of a single probabilistic value function. We subsequently demonstrate that the complexity of these computations is governed by three cardinal properties of the value function — (i) *monotonicity*, (ii) *submodularity*, and (iii) *supermodularity* — which together constitute foundational pillars of *combinatorial optimization*. Our analysis surfaces a number of counterintuitive phenomena concerning the manifestation of these properties across the explanatory regimes under consideration. In particular, whereas the *local* value functions exhibit neither monotonicity nor submodularity/supermodularity in any form, we establish that their *global* counterparts do in fact possess precisely these properties. This dichotomy affords us the leverage to prove a sequence of hitherto unavailable polynomial-time results for computing explanations endowed with provable guarantees in the global explainability regime, spanning an array of ML models that traverse the interpretability spectrum, such as neural networks, decision trees, and tree ensembles. Conversely, we show that even markedly simplified renditions of these very explanations become NP-hard to compute within the corresponding local explainability regime.

1 INTRODUCTION

Notwithstanding the considerable progress achieved in devising methods for explaining the decisions of ML models, a recurrent and regrettable finding throughout the literature is that many of the most desirable explanation types — those furnishing distinct provable guarantees — prove computationally intractable to obtain (Barceló et al., 2020; Van den Broeck et al., 2022), with this intractability characteristically intensifying in complex or markedly non-linear models (Barceló et al., 2020; Adolfi et al., 2025; 2024). Consequently, the *computational complexity* of obtaining explanations has emerged as a central theoretical preoccupation, with a proliferation of recent contributions endeavoring to delineate precisely which explanation types admit efficient computation for which classes of models, and which, by contrast, remain fundamentally beyond reach (Barceló et al., 2020; Wäldchen et al., 2021; Arenas et al., 2022; 2023; Marzouk & De La Higuera, 2024; Ordyniak et al., 2023; Laber, 2024; Bhattacharjee & Luxburg, 2024; Blanc et al., 2021; 2022).

From sufficient to contrastive reasons. Within the corpus of research addressing the computational complexity of generating explanations, two foundational classes of explanations for ML models have been subjected to particularly extensive scrutiny: (i) *sufficient reasons* and (ii) *contrastive reasons* (Barceló et al., 2020; Arenas et al., 2022; Audemard et al., 2022a; Arenas et al., 2021; Barceló et al., 2025; Marques-Silva & Ignatiev, 2022; Ignatiev et al., 2020b; Darwiche & Hirth, 2020). A *sufficient reason* constitutes a subset of input features S whose fixation to prescribed values guarantees that the model’s prediction persists unaltered, irrespective of whatever assignment is

imposed upon the complementary set $\bar{S}$. A *contrastive reason*, conversely, constitutes a subset of input features S whose modification occasions a change in the model’s prediction.

In contradistinction to additive attribution methods, which apportion importance scores across features yet frequently resist human interpretation (Kumar et al., 2020) or forfeit actionability (Bilodeau et al., 2024), sufficient and contrastive reasons furnish *discrete, condition-like explanations* that respond directly to the questions “what is enough to justify this prediction?” or “what must change to flip it?”. It is precisely their intuitive character that has secured them a central role across a multitude of classical XAI methods (Ribeiro et al., 2018; Carter et al., 2019; Ignatiev et al., 2019; Dhurandhar et al., 2018), that has been shown to confer greater efficacy in enhancing human prediction relative to additive models (Ribeiro et al., 2018; Yin & Neubig, 2022; Dhurandhar et al., 2018), and that has rendered them serviceable for an array of downstream tasks, including bias detection (Balkir et al., 2022; Carter et al., 2021; La Malfa et al., 2021; Muthukumar et al., 2018), model debugging (Jacovi et al., 2021), and anomaly detection (Davidson et al., 2025). A well-entrenched tenet further maintains that *smaller* sufficient and contrastive explanations enhance interpretability, thereby elevating *minimality* to the status of a central guarantee of interest (Ribeiro et al., 2018; Lopardo et al., 2023; Carter et al., 2019; Barceló et al., 2020; Arenas et al., 2022; Blanc et al., 2021; Wäldchen et al., 2021).

Barceló et al. (2020) undertook one of the earliest systematic investigations into the complexity of deriving sufficient and contrastive reasons. Their work established that ascertaining the minimal-sized sufficient reason for a decision tree is NP-hard, an intractability that escalates to Σ_2^P -hardness in the case of neural networks. As regards contrastive reasons, they demonstrated that computing the smallest admissible explanation is solvable in polynomial time for decision trees, yet reverts to NP-hardness for neural networks. Analogous hardness results were subsequently shown to obtain for tree ensembles as well (Izza & Marques-Silva, 2021; Ordyniak et al., 2024; Audemard et al., 2022b).

From non-probabilistic to probabilistic explanations. A recurrent objection levelled against the classical formulation of sufficient and contrastive reasons concerns their rigidity and want of flexibility: because they must hold in an absolute sense across entire domains, they are liable to engender explanations that are excessively large or uninformative (Ignatiev et al., 2019; 2020a; Arenas et al., 2022; Wäldchen et al., 2021). In order to redress these limitations, the literature has gravitated towards a more general formulation that inaugurates a *probabilistic* perspective on sufficient and contrastive reasons (Wäldchen et al., 2021; Arenas et al., 2022; Blanc et al., 2021; Izza et al., 2023; Xue et al., 2023). Under this framework, the objective becomes that of identifying subsets of input features that govern a prediction with probability exceeding a prescribed threshold δ .

Wäldchen et al. (2021) were the first to interrogate the complexity of probabilistic sufficient reasons, establishing that for CNF classifiers the problem is NP^{PP} -hard — an intractability that propagates to tree ensembles and neural networks (Barceló et al., 2020; Ordyniak et al., 2023). A pivotal theoretical insight arising in the probabilistic setting concerns the *lack of monotonicity* of the probability function (Arenas et al., 2022; Izza et al., 2023), a deficiency that renders even *subset minimal* explanations computationally hard to obtain. Strikingly, (Arenas et al., 2022) demonstrate that computing a subset minimal probabilistic sufficient reason is NP-hard even for decision trees — in marked contrast to the non-probabilistic case, wherein such explanations are computable in polynomial time (Huang et al., 2021; Barceló et al., 2020).

From local to global explanations. In a more recent contribution, Bassan et al. (2024) extend the complexity analysis from the *local* (non-probabilistic) setting — wherein explanations are tethered to individual predictions — to the *global* (non-probabilistic) setting, which seeks sufficient or contrastive reasons holding across entire domains. Yet, in common with other non-probabilistic methods (Ignatiev et al., 2019; Barceló et al., 2020; Arenas et al., 2021; Darwiche & Hirth, 2020), the attendant criteria are exceedingly stringent — arguably more so in the global regime than in the local one. For instance, any feature excluded from a global sufficient reason is thereby deemed strictly redundant (Bassan et al., 2024), a stipulation that frequently compels the explanation to encompass nearly all input features and thus to forfeit much of its informativeness.

OUR CONTRIBUTIONS

1. We consolidate a diverse array of previously disparate explanation-computation problems — encompassing the sufficient and contrastive, the probabilistic and non-probabilistic, and

the local and global variants alike — into a single unifying framework, formulated as a minimization task over a unified value function. We then isolate three fundamental properties of the value function that decisively govern the complexity of this task: (a) *monotonicity*, (b) *supermodularity*, and (c) *submodularity*.

2. Intriguingly, we establish that these properties comport themselves in *strikingly divergent manners* contingent upon the structure of the value function. In particular, we demonstrate the counterintuitive result that, whereas the *local* value functions for both sufficient and contrastive reasons are *non-monotonic*, their *global* counterparts are *monotonic non-decreasing*. Beyond this, we uncover additional properties peculiar to the global setting: the global sufficient value function proves *supermodular*, while the global contrastive value function proves *submodular* — in stark opposition to the local setting, wherein neither property obtains.
3. We harness these properties to derive novel complexity results for explanation computation, thereby bringing to light the intriguing finding that global explanations furnished with guarantees admit efficient computation, notwithstanding the computational hardness that persists for their local counterparts. We substantiate these findings across three widely adopted model types that traverse the interpretability spectrum: (i) neural networks, (ii) decision trees, and (iii) tree ensembles. We first prove that, whereas computing a subset-minimal local sufficient/contrastive probabilistic explanation is NP-hard even for decision trees (Arenas et al., 2022), its global counterpart is computable in polynomial time. We further extend this result to any black-box model — including complex models such as neural networks and tree ensembles — under empirical distributions. Specifically, we show that obtaining a subset-minimal global sufficient/contrastive explanation is attainable in polynomial time, whereas the local variant remains NP-hard for these very models.
4. Finally, we advance an even more forceful complexity result by exploiting the submodular and supermodular properties of the value functions in the global setting — properties conspicuously absent in the local case. Specifically, we show that provable *constant-factor* approximation guarantees are attainable for computing cardinally minimal global explanations — even for complex models such as neural networks or tree ensembles — once the empirical distribution is fixed. In sharp contradistinction, we establish strong inapproximability results for the local setting, demonstrating that no *bounded* approximation is achievable, even under exceedingly simple assumptions.

Owing to constraints of space, we confine the main text to an overview of our principal theorems and corollaries, relegating the complete proofs to the appendix.

2 PRELIMINARIES

Setting. We posit an input space of dimension $n \in \mathbb{N}$, populated by input vectors $\mathbf{x} := (\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n)$. Each coordinate $\mathbf{x}_i$ ranges over its associated domain $\mathcal{X}_i$, which may be either discrete or continuous, whence the complete input feature space is constituted as the product $\mathbb{F} := \mathcal{X}_1 \times \mathcal{X}_2 \times \dots \times \mathcal{X}_n$. Our attention is directed to classification models $f : \mathbb{F} \rightarrow [c]$, where $c \in \mathbb{N}$ denotes the number of classes, and we further presuppose a *generic* distribution $\mathcal{D} : \mathbb{F} \rightarrow [0, 1]$ over the input space. In numerous settings, however, an accurate approximation of $\mathcal{D}$ proves computationally infeasible; a natural expedient is therefore to operate instead over a fixed empirical dataset $\mathbf{D} := \{\mathbf{z}^1, \mathbf{z}^2, \dots, \mathbf{z}^{|\mathbf{D}|}\}$, which serves as a pragmatic proxy for the “true” distribution — obtained, for instance, by taking $\mathbf{D}$ to be a sampled subset of the available training data.

The explanations under investigation are of two kinds, local or global. In the local case, an explanation is directed at a particular prediction $\mathbf{x} \in \mathbb{F}$, furnishing a form of justification for why the model rendered the prediction $f(\mathbf{x})$ on that specific instance. In the global case, by contrast, an explanation aspires to capture the general rationale underlying the behavior of f across a more expansive region of the input space — independent of any particular $\mathbf{x}$ — and thereby to characterize its overarching decision-making logic.

Models. Although many of the results presented herein are of a general character (e.g., inherent properties of the value functions), we additionally furnish a number of model-specific complexity results for widely deployed ML models. With a view to spanning the interpretability spectrum in its

breadth, we have elected to analyze models ranging from those conventionally deemed “black-box” to those customarily regarded as “interpretable”. Concretely, our focus rests upon: (i) decision trees, (ii) neural networks (all architectures at least as expressive as feed-forward ReLU networks), and (iii) tree ensembles, encompassing both majority-voting ensembles (e.g., random forests) and weighted-voting ensembles (e.g., XGBoost). The formal definitions are deferred to the Appendix.

Distributions. We underscore that, in probabilistic explanation settings, the complexity may fluctuate appreciably with the input distribution $\mathcal{D}$. In particular, our attention is confined to three distributional regimes: (i) *general distributions*, which impose no specific assumptions upon $\mathcal{D}$ and hence subsume all conceivable distributions — these we invoke chiefly in the proofs of properties that hold universally; (ii) *empirical distributions*, which comprise all distributions induced by the finite dataset $\mathbf{D}$, an approach ubiquitously employed in XAI (Lundberg & Lee, 2017; Van den Broeck et al., 2022); and (iii) *independent distributions*, which posit that the features in $\mathcal{D}$ are mutually independent — a further assumption widely adopted throughout the XAI literature (Arenas et al., 2022; 2023; Lundberg & Lee, 2017; Ribeiro et al., 2018). It bears emphasizing that empirical distributions do not perforce entail feature independence; on the contrary, they are capable of encoding intricate dependencies extracted from finite datasets (Van den Broeck et al., 2022). The complete formal definitions of these distributions are supplied in the Appendix.

3 FORMS OF REASONS

In the present section, we introduce the explanation types under examination in this work, commencing with the stringent non-probabilistic definitions of (local/global) sufficient and contrastive reasons, and proceeding thereafter to their more flexible and readily generalizable probabilistic counterparts.

3.1 Non-Probabilistic SUFFICIENT AND CONTRASTIVE REASONS

Sufficient reasons. In the setting of *feature selection*, practitioners routinely single out the top k features that contribute to a model’s decision. We interrogate the well-established *sufficiency* criterion governing such a selection, a criterion consonant with a range of commonly employed explainability methods (Ribeiro et al., 2018; Carter et al., 2019; Ignatiev et al., 2019; Dasgupta et al., 2022). This feature selection may be prosecuted either locally — with attention trained upon a single prediction — or globally — across the entirety of the input domain. Adhering to standard conventions, we define a *local sufficient reason* as a subset of features $S \subseteq \{1, \dots, n\}$ such that, once the features in S are fixed to their corresponding values in $\mathbf{x}$, the model’s prediction persists at $f(\mathbf{x})$ irrespective of whatever values are assigned to the residual features $\bar{S}$. Formally, S constitutes a local sufficient reason for $\langle f, \mathbf{x} \rangle$ iff the following condition is satisfied:

$$\forall \mathbf{z} \in \mathbb{F}, \quad f(\mathbf{x}_S; \mathbf{z}_{\bar{S}}) = f(\mathbf{x}). \quad (1)$$

Here, $(\mathbf{x}_S; \mathbf{z}_{\bar{S}})$ designates a vector wherein the features in S inherit their values from $\mathbf{x}$, while those in $\bar{S}$ inherit theirs from $\mathbf{z}$. Local sufficient reasons stand in close kinship with *semi-factual* explanations (Kenny & Keane, 2021; Alfano et al., 2025), which seek alternative inputs $\mathbf{z}'$ that preserve the prediction unaltered (i.e., $f(\mathbf{z}') = f(\mathbf{x})$) while lying as “close” as possible to the original point. Upon instantiating $\mathbf{z}'$ as $(\mathbf{x}_S; \mathbf{z}_{\bar{S}})$ and gauging proximity through the Hamming distance, the two notions are seen to coincide.

A *global sufficient reason* (Bassan et al., 2024), in turn, is a subset of input features $S \subseteq \{1, \dots, n\}$ that discharges the role of a local sufficient reason for *every* admissible input $\mathbf{x} \in \mathbb{F}$:

$$\forall \mathbf{x}, \mathbf{z} \in \mathbb{F}, \quad f(\mathbf{x}_S; \mathbf{z}_{\bar{S}}) = f(\mathbf{x}). \quad (2)$$

Contrastive reasons. A further prevalent avenue for furnishing explanations proceeds by pinpointing subsets of input features whose alteration *modifies* a prediction (Dhurandhar et al., 2018; Mothilal et al., 2020; Guidotti, 2024). Explanations of this ilk aim to ascertain the minimal changes requisite to overturn a prediction. Formally, a subset $S \subseteq \{1, \dots, n\}$ is designated a *local contrastive reason* with respect to $\langle f, \mathbf{x} \rangle$ iff:

$$\exists \mathbf{z} \in \mathbb{F}, \quad f(\mathbf{z}_S; \mathbf{x}_{\bar{S}}) \neq f(\mathbf{x}). \quad (3)$$

Contrastive reasons are, moreover, intimately bound up with *counterfactual* explanations (Guidotti, 2024; Mothilal et al., 2020), which seek a proximate assignment $\mathbf{x}'$ upon which the prediction is inverted, i.e., $f(\mathbf{x}') \neq f(\mathbf{x})$. Should the distance metric be taken to be the Hamming weight, the two notions once again coincide.

In analogy with sufficient reasons, one may likewise seek a subset of input features whose alteration overturns the prediction for all inputs residing within the domain of interest. This form of explanation is, in addition, closely allied to approaches for the identification of bias (Arenas et al., 2021; Bassan et al., 2024; Darwiche & Hirth, 2020), as well as to *group* counterfactual explanation methods that seek counterfactuals across multiple data instances (Carrizosa et al., 2024; Warren et al., 2024). Formally, we designate a subset S a *global contrastive reason* with respect to f iff:

$$\forall \mathbf{x} \in \mathbb{F}, \quad \exists \mathbf{z} \in \mathbb{F}, \quad f(\mathbf{z}_S; \mathbf{x}_{\bar{S}}) \neq f(\mathbf{x}). \quad (4)$$

3.2 Probabilistic SUFFICIENT AND CONTRASTIVE REASONS

As adumbrated in the introduction, non-probabilistic sufficient and contrastive reasons are frequently faulted for imposing conditions of excessive stringency, a shortcoming that has impelled a migration towards *probabilistic* definitions (Arenas et al., 2022; Ribeiro et al., 2018; Izza et al., 2023; Wäldchen et al., 2021; Bounia & Koriche, 2023; Subercaseaux et al., 2025; Wang et al., 2021a; Blanc et al., 2021), which generalize these requirements by demanding merely that the guarantees hold with probability at least $\delta \in [0, 1]$. The special case $\delta = 1$ recovers, as a degenerate instance, the original non-probabilistic definitions.

Probabilistic sufficient reasons. We define $S \subseteq \{1, \dots, n\}$ to be a *local δ -sufficient reason* with respect to $\langle f, \mathbf{x} \rangle$ if, upon fixing the features in S to their corresponding values in $\mathbf{x}$ and sampling the residual features from a distribution $\mathcal{D}$, the classification is preserved with probability at least δ . Stated otherwise:

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S) \geq \delta. \quad (5)$$

where $\mathbf{z}_S = \mathbf{x}_S$ signifies that the features in S of the vector $\mathbf{z}$ are fixed to their corresponding values in $\mathbf{x}$. Embracing the standard conception of global explanations — namely, averaging over all inputs within the global domain — we define a subset $S \subseteq \{1, \dots, n\}$ to be a *global δ -sufficient reason* with respect to $\langle f \rangle$ if the expectation of the local sufficiency probability, taken over samples drawn from the distribution $\mathcal{D}$, attains a value of at least δ . Stated otherwise:

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \geq \delta. \quad (6)$$

Probabilistic contrastive reasons. In parallel with the non-probabilistic case, we define a *local δ -contrastive reason* as a subset of input features whose variation overturns a prediction with a certain probability. Here, in contradistinction to sufficient reasons, we fix the features of the *complementary* set $\bar{S}$ to their respective values in $\mathbf{x}$, and, upon permitting the features in S to vary, we demand that the prediction diverge from the original prediction $f(\mathbf{x})$ with probability exceeding δ . Formally:

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}}) \geq \delta. \quad (7)$$

For the global setting, and in analogy with global sufficient reasons, we define a subset S to be a *global δ -contrastive reason* by taking the expectation of the local contrastive probability over samples drawn from the distribution $\mathcal{D}$ and requiring that this expected value exceed δ :

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}})] \geq \delta. \quad (8)$$

4 FORMS OF MINIMALITY

As foreshadowed in the introduction, across all the explanation forms canvassed hitherto — be they sufficient or contrastive, local or global — a prevailing assumption throughout the literature holds that explanations of *smaller size* carry greater meaning, thereby investing their *minimality* with the status of a particularly consequential provable guarantee. In the present study, we examine two central notions of minimality across the full range of our explanation types:

Definition 1. Assuming a subset $S \subseteq \{1, \dots, n\}$ is an explanation, then:

1. S is a *cardinally-minimal explanation* (Barceló et al., 2020; Bassan et al., 2024) iff S has the smallest explanation cardinality $|S|$ (i.e., there is no explanation S' such that $|S'| < |S|$).
2. $S \subseteq \{1, \dots, n\}$ is a *subset-minimal explanation* (Arenas et al., 2022; Ignatiev et al., 2019) iff S is an explanation, and any $S' \subsetneq S$ is not an explanation.

We observe that cardinal-minimality is strictly the more exacting of the two: every cardinally minimal S is perforce subset-minimal, though the converse fails to hold (see Appendix B.3 for an illustrative example). We deploy the terminology of subset and cardinal minimality, in preference to that of local and global minima, so as to forestall any conflation with local versus global explanations (that is, input-level versus domain-level reasoning). Both notions are applicable to *all* explanation forms we analyze. For instance, a cardinally minimal local probabilistic δ sufficient reason is that possessing the smallest $|S|$, whereas a subset-minimal one is any S for which no proper subset $S' \subsetneq S$ likewise qualifies.

5 A UNIFIED COMBINATORIAL OPTIMIZATION TASK

Remarkably, all the computational problems adduced hitherto — local or global, sufficient or contrastive, probabilistic or otherwise — admit a common recasting as the search for a minimal-size subset S satisfying $v(S) \geq \delta$, where in the non-probabilistic settings one simply sets $\delta := 1$. We now introduce the requisite notation for the value functions: let v_{suff}^{ℓ} denote the local sufficiency probability of equation 5, namely $\Pr_{\mathbf{z} \sim \mathcal{D}_p}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)$, and let v_{suff}^g denote its global variant (equation 6). In like manner, let v_{con}^{ℓ} and v_{con}^g denote the local and global contrastive probabilities (equations 7 and 73, respectively). Equipped with this notation, we now formally articulate the unified task of computing a cardinally minimal δ -local/global sufficient/contrastive reason:

Cardinally Minimal δ -Explanation:

Input: Model f , a distribution $\mathcal{D}$, (possibly, an input $\mathbf{x}$), a *general* value function $v : 2^n \rightarrow [0, 1]$ (defined using f , $\mathcal{D}$, and possibly $\mathbf{x}$), and some $\delta \in [0, 1]$.

Output: A subset $S \subseteq [n]$ such that $v(S) \geq \delta$ and $|S|$ is minimal.

In parallel fashion, for the relaxed condition wherein the objective is to procure a subset-minimal rather than a cardinally-minimal local or global sufficient/contrastive reason, we posit the following relaxed optimization objective:

Subset Minimal δ -Explanation:

Input: Model f , a distribution $\mathcal{D}$, (possibly, an input $\mathbf{x}$), a *general* value function $v : 2^n \rightarrow [0, 1]$ (defined using f , $\mathcal{D}$, and possibly $\mathbf{x}$), and some $\delta \in [0, 1]$.

Output: A subset $S \subseteq [n]$ such that $v(S) \geq \delta$ and for any $S' \subsetneq S$ it holds that $v(S') < \delta$.

Properties of v that affect the complexity. We now delineate several cardinal properties of the value function which, as we subsequently demonstrate, play a decisive role in dictating the complexity of generating explanations. The first such property is *non-decreasing monotonicity*, which guarantees that the marginal contribution $v(S \cup \{i\}) - v(S)$ remains uniformly non-negative. Formally:

Definition 2. We say that a value function v maintains non-decreasing monotonicity if for any $S \subseteq \{1, \dots, n\}$ and any $i \in \{1, \dots, n\}$ it holds that: $v(S \cup \{i\}) \geq v(S)$.

The remaining cardinal properties, *supermodularity* and *submodularity*, concern the manner in which the marginal contribution $v(S \cup \{i\}) - v(S)$ evolves. Specifically, in the supermodular case this contribution constitutes a monotone non-decreasing function of the underlying set, whereas under the dual definition of the *submodular* case the marginal contribution $v(S \cup \{i\}) - v(S)$ constitutes, conversely, a monotone non-increasing function. Formally:

Definition 3. Let there be some value function v , some $S \subseteq S' \subseteq \{1, \dots, n\}$, and $i \notin S'$. Then:

1. v maintains supermodularity iff it holds that: $v(S \cup \{i\}) - v(S) \leq v(S' \cup \{i\}) - v(S')$.
2. v maintains submodularity iff it holds that: $v(S \cup \{i\}) - v(S) \geq v(S' \cup \{i\}) - v(S')$.

6 UNRAVELING NEW PROPERTIES OF THE GLOBAL VALUE FUNCTIONS

Prior work has established that, in the local setting of non-probabilistic explanations, subset-minimal sufficient or contrastive reasons are amenable to computation by virtue of monotonicity — a property that obtains, however, solely in the restricted case $\delta = 1$. This restriction is severely confining and forfeits practical flexibility. While one might aspire to generalize to probabilistic guarantees for arbitrary δ , prior results demonstrate that monotonicity founders in this more general setting, whereby the computation of explanations is rendered commensurately more onerous (Arenas et al., 2022; Kozachinskiy, 2023; Izza et al., 2023; Subercaseaux et al., 2025; Izza et al., 2024b).

In the yet more extreme case of the *global* and *non-probabilistic* setting, Bassan et al. (2024) established the stringent *uniqueness* property — namely, that there exists exactly *one* subset-minimal explanation. The requirement that $\delta = 1$, however, renders this setting acutely restrictive, above all in the global case, wherein the explanation conditions must hold for *all* inputs. Indeed, Bassan et al. (2024) proves that this unique minimal subset coincides with the subset comprising all features that are not strictly redundant (which may, in practice, amount to all of them). We prove that in the general global *probabilistic* setting — for arbitrary δ — this uniqueness property in fact does *not* hold, and that the number of such subsets may even grow *exponentially*.

Proposition 1. *While the non-probabilistic case ($\delta = 1$) admits a unique subset-minimal global sufficient reason (Bassan et al., 2024), in the general probabilistic setting (for arbitrary δ), there exist functions f that have $\Theta(\frac{2^n}{\sqrt{n}})$ subset-minimal global sufficient reasons.*

Intriguingly, although the uniqueness property fails to obtain in the general case for arbitrary δ (being confined to the special case $\delta = 1$), we show that the crucial *monotonicity* property does hold for the global value function across the entire range of δ . This holds for both the sufficient (v_{suff}^g) and contrastive (v_{con}^g) value functions alike. The finding is a surprising one, inasmuch as it stands in sharp opposition to the local setting, wherein the corresponding value functions (v_{suff}^ℓ and v_{con}^ℓ) fail to satisfy this very property:

Proposition 2. *While the local probabilistic setting (for any δ) lacks monotonicity — i.e., the value functions v_{con}^ℓ and v_{suff}^ℓ are non-monotonic (Arenas et al., 2022; Izza et al., 2023; Subercaseaux et al., 2025; Izza et al., 2024b) — in the global probabilistic setting (also for any δ), both value functions v_{con}^g and v_{suff}^g are monotonic non-decreasing.*

Beyond the surprising insight that monotonicity holds for the global value functions — yet fails for their local counterpart — we discern additional structural properties peculiar to the global setting, namely submodularity and supermodularity. In particular, we show that, under the commonplace assumption of feature independence, the global sufficient value function v_{suff}^g exhibits supermodularity. Its local counterpart v_{suff}^ℓ , by contrast, fails to exhibit this property even under the considerably more restrictive assumption of a uniform (and hence independent) input distribution. Specifically:

Proposition 3. *While the local probabilistic sufficient setting (for any δ) lacks supermodularity — even when $\mathcal{D}$ is uniform, i.e., the value function v_{suff}^ℓ is not supermodular — in the global probabilistic setting (also for any δ), when $\mathcal{D}$ exhibits feature independence, the value function v_{suff}^g is supermodular.*

Intriguingly, for the second family of explanation settings — specifically, that of computing a global probabilistic *contrastive* reason — we show that the corresponding value function is not supermodular but is instead *submodular*. This result proves especially surprising when set against the local setting, wherein the value function is neither submodular nor supermodular.

Proposition 4. *While the local probabilistic contrastive setting (for any δ) lacks submodularity — even when $\mathcal{D}$ is uniform, i.e., the value function v_{con}^ℓ is not submodular — in the global probabilistic setting (also for any δ), when $\mathcal{D}$ exhibits feature independence, the value function v_{con}^g is submodular.*

7 COMPUTATIONAL COMPLEXITY RESULTS

7.1 SUBSET MINIMAL EXPLANATIONS

In this section, we scrutinize the complexity of computing *subset minimal* explanations (local/global, sufficient/contrastive) across the several model types under analysis. The property of decisive

importance here is the *monotonicity* of the value function. The preceding section established that monotonicity holds for both global value functions, v_{con}^g and v_{suff}^g , yet fails for the local value functions, v_{con}^l and v_{suff}^l . This dichotomy is instrumental in demonstrating that a greedy algorithm converges to a subset minimal explanation in the global setting while faltering in the local one. Accordingly, we shall bring to light the surprising finding that computing the various local subset-minimal explanation forms is hard, whereas computing their subset-minimal global counterparts is tractable (that is, polynomial-time solvable). We begin by introducing the following generalized greedy algorithm:

Algorithm 1 Subset Minimal Explanation Search

Input Value function v , and some $\delta \in [0, 1]$

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1:  $S \leftarrow \{1, \dots, n\}$ 
2: while  $\min_{i \in S} v(S \setminus \{i\}) \geq \delta$  do
3:    $j \leftarrow \operatorname{argmax}_{i \in S} v(S \setminus \{i\})$ 
4:    $S \leftarrow S \setminus \{j\}$ 
5: end while
6: return  $S$ 

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$\triangleright S$ is a (subset minimal?) δ -explanation

Algorithm 1 is designed to procure a subset-minimal δ -explanation. We initialize the algorithm with the subset S set to the entire input space $\{1, \dots, n\}$. At each iteration, we ascertain whether the minimal value attainable by $v(S \setminus \{i\})$ exceeds δ , and, if so, we excise from S the feature j that occasions the least decrease in the value function — that is, the feature j maximizing $v(S \setminus \{j\})$. Upon the termination of this iterative procedure, we return S .

The decisive determinant of whether Algorithm 1 yields a subset-minimal explanation is the *monotonicity* property of the value function v . The algorithm terminates in a phase wherein the removal of any individual feature from S renders $v(S \setminus \{i\})$ smaller than δ ; monotonicity, however, guarantees that this holds for any $v(S \setminus S')$, thereby furnishing a substantially stronger guarantee. In light of the monotonicity property of the value functions established in the foregoing sections, we derive the following proposition:

Proposition 5. *Computing Algorithm 1 with the local value functions v_{con}^l and v_{suff}^l does not always converge to a subset minimal δ -sufficient/contrastive reason. However, computing it with the global value functions v_{con}^g or v_{suff}^g necessarily produces subset minimal δ -sufficient/contrastive reasons.*

Building upon this result, we now proceed to establish new complexity findings for deriving the various forms of subset-minimal explanations within our framework, with due regard to the several model types under analysis.

Decision trees. We commence by examining a highly simplified and ostensibly “interpretable” scenario — specifically, one in which the model f is a decision tree and the distribution $\mathcal{D}$ is independent. Within this simplified setting, we exhibit a strict separation: Arenas et al. (2022) established the surprising intractability result that, unless $\text{PTIME}=\text{NP}$, no polynomial-time algorithm exists for computing a subset-minimal local δ -sufficient reason for decision trees under independent distributions (indeed, even under the uniform distribution). We demonstrate, by contrast, the unexpected result that this very problem admits efficient solution in the global setting — that is, a subset-minimal global δ -sufficient reason for decision trees can in fact be computed in polynomial time. Formally:

Theorem 1. *If f is a decision tree and the probability term v_{suff}^g can be computed in polynomial time given the distribution $\mathcal{D}$ (which holds for independent distributions, among others), then obtaining a subset-minimal global δ -sufficient reason can be obtained in polynomial time. However, unless $\text{PTIME}=\text{NP}$, no polynomial-time algorithm exists for computing a local δ -sufficient reason for decision trees even under independent distributions.*

Extension to other tractable models. We further remark that an analogous tractability guarantee for computing global explanations obtains for *orthogonal DNFs* (Crama & Hammer, 2011) as well — which, we briefly recall, are Disjunctive Normal Form formulas whose terms are pairwise mutually exclusive. Establishing the result for this class is of value precisely because orthogonal DNFs *subsume* decision trees while retaining their clean structural properties, thereby permitting our guarantees to extend beyond tree-structured models. For the sake of completeness, we furnish the full argument in Appendix N.

Neural networks, tree ensembles, and other complex models. We now extend our foregoing results to more complex models lying beyond decision trees. Specifically, we shall demonstrate that whenever the distribution $\mathcal{D}$ is derived from empirical distributions — and under the elementary assumption that the model f admits polynomial-time inference — it follows that computing a subset-minimal global δ -sufficient or contrastive reason is achievable in polynomial time.

Proposition 6. *For any model f , and empirical distribution $\mathcal{D}$ — computing a subset-minimal global δ -sufficient or δ -contrastive reason for f can be done in polynomial time.*

This forceful complexity outcome, holding as it does for *any* model under an empirical-distribution assumption, enables us to draw a still finer distinction between the complexity of the local and global explanation settings. Specifically, for certain complex models, computing subset-minimal *local* explanations remains intractable even when the analysis is confined to empirical distributions. We establish this fact for both neural networks and tree ensembles, which culminates in the following theorem asserting a strict complexity separation:

Theorem 2. *Assuming f is a neural network or a tree ensemble, and $\mathcal{D}$ is an empirical distribution — there exist polynomial-time algorithms for obtaining subset minimal global δ -sufficient and contrastive reasons. However, unless $\text{PTIME}=\text{NP}$, there is no polynomial time algorithm for computing a subset minimal local δ -sufficient reason or a subset minimal local δ -contrastive reason.*

7.2 APPROXIMATE CARDINALLY MINIMAL EXPLANATIONS

In this subsection, we redirect our attention from subset-minimal sufficient/contrastive reasons to the still more formidable task of computing a *cardinally minimal* δ -sufficient/contrastive reason. We shall demonstrate that, in the global setting, the interplay between the supermodularity/submodularity and the monotonicity of the value function equips us to establish novel *provable approximations* for computing explanations, whereas the computation of these same explanations in the local setting remains intractable. This result serves to further reinforce the surprising disparity between the tractability of computing global explanations and that of their local counterparts.

A unified greedy approximation algorithm. When one operates with a non-decreasing monotone submodular function, the problem of identifying a cardinally minimal explanation aligns closely with the *submodular set cover* problem (Wolsey, 1982); the employment of a *supermodular* function, by contrast, engenders a non-submodular variant of this problem (Shi et al., 2021). Both problems have attracted considerable attention on account of their strong approximation guarantees (Wolsey, 1982; Iyer & Bilmes, 2013; Chen & Crawford, 2023). In the context of submodular set cover optimization, the canonical approach rests upon a classical greedy algorithm, which we first delineate (Algorithm 2). This algorithm furnishes the foundation for approximating a cardinally minimal sufficient or contrastive δ -reason, and we shall subsequently examine the specific guarantees it affords in each of the two cases.

Algorithm 2 Cardinally Minimal Explanation Approximation Search

Input Value function v , and some $\delta \in [0, 1]$

```

1:  $S \leftarrow \emptyset$ 
2: while  $\max_{i \in S} v(S \cup \{i\}) < \delta$  do
3:    $j \leftarrow \operatorname{argmax}_{i \in S} v(S \cup \{i\})$ 
4:    $S \leftarrow S \cup \{j\}$ 
5: end while
6: return  $S$   $\triangleright |S|$  is a (provable approximation?) of a cardinally minimal  $\delta$ -explanation

```

Algorithm 2 bears a close resemblance to Algorithm 1, save that it proceeds bottom-up rather than top-down. It is initialized with an empty subset and augments it incrementally, at each step adjoining the feature that minimizes the increase in $v(S \cup i)$, and terminating once the addition of any further feature would drive the value past δ .

Cardinally minimal contrastive reasons. We begin with global contrastive reasons, for which monotonicity and submodularity both obtain, whereby the task reduces to the classical *submodular set cover* problem and we are enabled to invoke Wolsey (1982)’s classical guarantee by way of Algorithm 2. For integer-valued objectives, the algorithm attains a Harmonic-based factor, and,

more generally, an $\mathcal{O}\left(\ln\left(\frac{v([n])}{\min_{i \in [n]} v(i)}\right)\right)$ -approximation. Under an *empirical* distribution $\mathcal{D}$ of fixed sample size, this reduces to a *constant approximation* — one merely *logarithmic* in the sample size — thereby yielding a substantially strong guarantee. In the *local* setting, by contrast, no *bounded* approximation exists even for a *single* sample point, which marks a pronounced chasm between the global and local cases.

Theorem 3. *Given a neural network or tree ensemble f and an empirical distribution $\mathcal{D}$ over a fixed dataset D , Algorithm 2 yields a constant $\mathcal{O}\left(\ln\left(\frac{v_{con}^g([n])}{\min_{i \in [n]} v_{con}^g(\{i\})}\right)\right)$ -approximation, bounded by $\mathcal{O}(\ln(|D|))$, for computing a global cardinally minimal δ -contrastive reason for f , assuming feature independence. In contrast, unless $PTIME=NP$, no bounded approximation exists for computing a local cardinally minimal δ -contrastive reason for any $\langle f, \mathbf{x} \rangle$, even when $|D| = 1$.*

We furthermore supply a matching *lower bound* for the bound of Theorem 3 in the specific case of an empirical distribution unencumbered by any independence assumption (see Appendix M).

Cardinally minimal sufficient reasons. In contradistinction to the submodular set cover problem — which is bound up with cardinally minimal global *contrastive* reasons and admits strong approximations — the supermodular variant, associated with global *sufficient* reasons, proves the more recalcitrant to approximate. It nonetheless affords guarantees whenever the function possesses bounded curvature (Shi et al., 2021). The *total curvature* of a function $v : 2^n \rightarrow \mathbb{R}$ is defined as $k^f := 1 - \min_{i \in [n]} \frac{v([n]) - v([n] \setminus i)}{v(i) - v(\emptyset)}$. Drawing upon results of Shi et al. (2021), we show that Algorithm 2 attains an $\mathcal{O}\left(\frac{1}{1-k^f} + \ln\left(\frac{v([n])}{\min_{i \in [n]} v(i)}\right)\right)$ -approximation.

Notably, under a fixed empirical distribution the approximation once again becomes constant. Whereas contrastive reasons admit the tighter $\mathcal{O}(\ln(|D|))$ bound, sufficient reasons incur an additional $\frac{1}{1-k^f}$ factor — yet still yield a constant-factor approximation. In sharp contrast, the local variant remains inapproximable, forfeiting any *bounded* approximation even when $|D| = 1$.

Theorem 4. *Given a neural network or tree ensemble f and an empirical distribution $\mathcal{D}$ over a fixed dataset D , Algorithm 2 yields a constant $\mathcal{O}\left(\frac{1}{1-k^f} + \ln\left(\frac{v_{suff}^g([n])}{\min_{i \in [n]} v_{suff}^g(\{i\})}\right)\right)$ -approximation for computing a global cardinally minimal δ -sufficient reason for f , assuming feature independence. In contrast, unless $PTIME=NP$, there is no bounded approximation for computing a local cardinally minimal δ -sufficient reason for any $\langle f, \mathbf{x} \rangle$, even when $|D| = 1$.*

Taken together, these findings fortify the conclusions of Subsection 7.1, which evidenced the tractability of computing subset-minimal global explanations in stark contrast to their local counterparts. Here we establish, in addition, that approximating cardinally minimal global explanations is likewise tractable — unlike their local counterparts, which prove inapproximable.

8 LIMITATIONS AND FUTURE WORK

While many of our most consequential findings — particularly those bearing upon the fundamental properties of value functions — hold in full generality, we also instantiate them so as to yield concrete complexity results for specific model classes (e.g., neural networks), distributional assumptions (e.g., empirical distributions), and explanation definitions within our framework. Other conceivable settings, it goes without saying, remain open to analysis. We contend, nonetheless, that our findings afford compelling insight into the foundational underpinnings of explanations, together with new tractability and intractability results, which jointly lay a robust foundation for the investigation of a broader compass of explainability scenarios in future work.

More specifically, our study brings into relief several salient open theoretical questions:

1. In the first place, it would be of interest to investigate whether certain of the tractable complexity results we have obtained for global explainability admit further tightening for simpler model classes, such as decision trees, linear models, or other models of inherently tractable structure. Although local explanations for such models are frequently intractable to compute (Arenas et al., 2022; Subercaseaux et al., 2025), the global explanation forms we study in this work exhibit strong structural properties that occasion markedly improved

complexity guarantees — guarantees that may perhaps be sharpened yet further. Tightening the approximation guarantees under empirical distributions constitutes, moreover, a key open direction.

2. In the second place, whenever the *exact* computation of expectations proves infeasible, it is natural to inquire whether alternative techniques — such as Fully Polynomial Randomized Approximation Schemes (FPRAS) for model classes like DNF formulas (Meel et al., 2019) (and hence for tree ensembles), or Monte Carlo-based approximations of the kind employed in (Subercaseaux et al., 2025) — might be brought to bear in approximating these expectations. A pivotal open challenge lies in ascertaining how the minimality guarantees may be preserved under approximation error, and what tolerance levels continue to permit the guarantees to hold.
3. Finally, we conjecture that the study of alternative notions of feature importance — for instance, alternative loss measures such as KL divergence (Conmy et al., 2023) — may reveal analogous patterns of monotonicity, submodularity, or supermodularity in the transition from local to global explanations. Properties of this kind could, in turn, enable efficient feature selection under these alternative metrics as well.

9 CONCLUSION

We advance a unified framework for the evaluation of diverse explanations and thereby lay bare a stark contrast between local and global sufficient and contrastive reasons. Notably, whereas the *local* explanation variants are bereft of any structural form endemic to combinatorial optimization — be it monotonicity, submodularity, or supermodularity — we prove that their *global* counterparts exhibit precisely these crucial properties: (i) *monotonicity*, (ii) *submodularity* in the case of contrastive reasons, and (iii) *supermodularity* for sufficient reasons. These proofs constitute the bedrock upon which we establish a series of surprising complexity results, demonstrating that global explanations endowed with provable guarantees admit efficient computation even for complex model classes such as neural networks. In sharp contradistinction, we prove that computing the corresponding local explanations remains NP-hard — even in highly simplified scenarios. Taken in the aggregate, our results uncover foundational properties of explanations and chart both the tractable and the intractable frontiers, thereby opening new avenues for future research.

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Appendix

The appendix contains formalizations and proofs that were mentioned throughout the paper:

Appendix A contains extended related work on Formal XAI.

Appendix B contains the formalizations of the models and distributions used in this work.

Appendix C contains the proof of Proposition 1.

Appendix D contains the proof of Proposition 2.

Appendix E contains the proof of Proposition 3.

Appendix F contains the proof of Proposition 4.

Appendix G contains the proof of Proposition 5.

Appendix H contains the proof of Theorem 1.

Appendix I contains the proof of Proposition 6.

Appendix J contains the proof of Theorem 2.

Appendix K contains the proof of Theorem 3.

Appendix L contains the proof of Theorem 4.

Appendix O contains an LLM usage disclosure.

A EXTENDED RELATED WORK ON FORMAL XAI

Our work is situated within the line of research on *formal explainable AI* (formal XAI) (Marques-Silva & Ignatiev, 2022), which investigates explanations endowed with provable guarantees (Yu et al., 2023; Darwiche & Ji, 2022; Darwiche, 2023; Shih et al., 2018; Azzolin et al., 2025; Audemard et al., 2021; Calautti et al., 2025). A central preoccupation of this area concerns the computational complexity of generating such guaranteed explanations (Barceló et al., 2020; ?; Marzouk et al., 2025a;b; Marzouk & De La Higuera, 2024; Bassan et al., 2025e; Amir et al., 2024; Blanc et al., 2021; 2022; Jaakkola et al., 2025; Amgoud & Cooper, 2026; Geibinger et al., 2025). Because the derivation of formal explanations is frequently computationally intractable for expressive models such as neural networks (Adolfi et al., 2024; Barceló et al., 2020) and tree ensembles (Ordyniak et al., 2024; Bassan et al., 2025a), a substantial portion of the literature has trained its attention upon more tractable model classes (Marques-Silva & Ignatiev, 2023), among them decision trees (Bounia, 2025; Arenas et al., 2022; Bounia, 2024; Bounia & Koriche, 2023), monotonic models (Marques-Silva et al., 2021; Harzli et al., 2023), linear models (Marques-Silva et al., 2020; Subercaseaux et al., 2025), and additive models (Bassan et al., 2026).

Beyond complexity-theoretic investigations, considerable effort has likewise been expended on the computation of formal explanations in practice by recourse to automated reasoning tools — such as MaxSAT and MILP solvers for tree ensembles (Audemard et al., 2023; 2022a; Ignatiev et al., 2022; Boumazouza et al., 2021), and neural network verification frameworks (Wang et al., 2021b; Wu et al., 2024a) for the derivation of certified explanations on neural networks (Wu et al., 2024b; Bassan & Katz, 2023; Bassan et al., 2023; Izza et al., 2024a; Hadad et al., 2026; Ignatiev et al., 2019). Additional approaches endeavor to alleviate the inherent computational burden by way of abstractions (De Palma et al., 2025; Bassan et al., 2025b; Boumazouza et al., 2026), relaxations such as smoothing (Xue et al., 2023; Anani et al., 2025; Jin et al., 2025), or training-time interventions (Alvarez Melis & Jaakkola, 2018; Bassan et al., 2025c;d).

Finally, within formal XAI, a related line of work examines methods that harness combinatorial and submodular optimization techniques to ameliorate the computation of *local* probabilistic sufficient reasons (Bounia, 2025; Bounia & Koriche, 2023), particularly for decision tree classifiers. As is noted both in that literature and in the present work, the value-function objective underlying these local explanation variants is not, in practice, submodular. Greedy algorithms inspired by submodular optimization nonetheless frequently serve as effective heuristics and can, empirically, yield high-quality explanations. Our work, by contrast, forges a rigorous theoretical connection between combinatorial and submodular optimization objectives on the one hand and the *global* variants of sufficient and contrastive reasons on the other — a connection that furnishes formal guarantees and opens the door to a broader spectrum of principled algorithmic implementations.

B MODEL AND DISTRIBUTION FORMALIZATIONS

In this section, we render precise the models and distributions invoked throughout the paper. Specifically, Subsection B.1 defines the model families, whereas Subsection B.2 formalizes the distributions.

B.1 MODEL FORMALIZATIONS

In this subsection, we render precise the three base-model types analyzed throughout the paper: (i) (axis-aligned) decision trees, (ii) linear classifiers, and (iii) neural networks with ReLU activations.

Decision Trees. We define a decision tree (DT) to be a directed acyclic graph representing a function $f : \mathbb{F} \rightarrow [c]$, where $c \in \mathbb{N}$ denotes the number of classes. The graph encodes the function in accordance with the following stipulations: (i) each internal node v is assigned a distinct binary input feature drawn from the set $\{1, \dots, n\}$; (ii) every internal node v possesses at most k outgoing edges, each corresponding to a value in $[k]$ assigned to v ; (iii) along any path α in the decision tree, each variable is encountered at most once; and (iv) each leaf node is labeled with a class from $[c]$. Consequently, an assignment of values to the inputs $\mathbf{x} \in \mathbb{F}$ uniquely determines a path α from the root to a leaf in the DT, wherein the function output $f(\mathbf{x})$ corresponds to a class label $i \in [c]$. The size of the DT, denoted $|f|$, is defined as the total number of edges in the graph. To accommodate flexible modeling, the ordering of the input variables $\{1, \dots, n\}$ may vary across distinct paths α and α' , subject to the proviso that no variable recurs along any single path.

Neural Networks. We present our *hardness* proofs for neural networks with ReLU activations, though our *tractability results* (namely, polynomial-time algorithms) apply to any architecture admitting polynomial-time inference — a standard assumption. Accordingly, every separation result between the tractable and intractable cases that we establish carries over to *any neural architecture* at least as expressive as a standard feed-forward ReLU network, thereby subsuming a great many widely deployed models such as ResNets, CNNs, Transformers, Diffusion models, and beyond. It bears noting, moreover, that any ReLU network is representable as a fully connected network upon assigning zero weights and biases to the absent connections. Adhering to standard conventions (Barceló et al., 2020; Bassan et al., 2024; Adolfi et al., 2025), we therefore assume the network to be fully connected. Concretely, our analysis pertains to multi-layer perceptrons (MLPs). Formally, an MLP f comprises $t - 1$ *hidden layers* ($g^{(j)}$ for $j = 1, \dots, t - 1$) together with a single output layer $g^{(t)}$, each layer being defined recursively as:

$$g^{(j)} := \sigma^{(j)}(g^{(j-1)}W^{(j)} + b^{(j)}) \quad (9)$$

where $W^{(j)}$ denotes the weight matrix, $b^{(j)}$ the bias vector, and $\sigma^{(j)}$ the activation function of the j -th layer. The model thus comprises, in total, t weight matrices ($W^{(1)}, \dots, W^{(t)}$), t bias vectors ($b^{(1)}, \dots, b^{(t)}$), and t activation functions ($\sigma^{(1)}, \dots, \sigma^{(t)}$).

The input layer is defined as $g^{(0)} = \mathbf{x} \in \{0, 1\}^n$, representing the binary input vector. The dimensions of the network are dictated by a sequence of positive integers $d_0, \dots, d_t$, with the weight matrices and bias vectors given, respectively, by $W^{(j)} \in \mathbb{Q}^{d_{j-1} \times d_j}$ and $b^{(j)} \in \mathbb{Q}^{d_j}$ — parameters that are learned during training. Inasmuch as the model operates as a binary classifier over n features, we set $d_0 = n$ and $d_t = 1$. The hidden layers employ the ReLU activation function, defined by $\text{ReLU}(x) = \max(0, x)$. Although a sigmoid activation is customarily used during training, for interpretability purposes we assume that the output layer applies a threshold-based step function, defined by $\text{step}(\mathbf{z}) = 1$ if $\mathbf{z} \geq 0$ and $\text{step}(\mathbf{z}) = 0$ otherwise.

Tree Ensembles. Although a profusion of popular ensemble methods exists, our objective being the provision of post-hoc explanations, we direct our attention to the *inference* phase rather than to the training process. Our analysis is centered upon ensemble families that rely, during inference, upon either *majority voting* (e.g., Random Forests) or *weighted voting* (e.g., XGBoost). As with our treatment of neural networks, however, our *tractability results* — namely, polynomial-time algorithms — extend to any conceivable ensemble configuration admitting polynomial-time inference, thereby subsuming an even more capacious range of ensemble configurations.

Majority Voting Inference. Under *majority voting* inference, the final prediction $f(\mathbf{x})$ is assigned to that class $j \in [c]$ which garners the majority of votes among the individual predictions $f_i(\mathbf{x})$ rendered by all $i \in [k]$ (that is, by each tree in the ensemble).

$$f(\mathbf{x}) := \text{majority}(\{f_i(\mathbf{x}) \mid i \in [k]\}) \quad (10)$$

where $\text{majority}(S)$ denotes the most frequently occurring label in the multiset S . In the event of a tie, resolution is effected by a fixed tie-breaking rule (e.g., lexicographic order or predefined priority).

Weighted Voting Inference. Under *weighted voting* inference, each model in the ensemble is assigned a weight $\phi_i \in \mathbb{Q}$ that encodes its relative importance, and the predicted class is that commanding the highest aggregate weight across all models. Formally, for any $\mathbf{x} \in \mathbb{F}$, we define f as:

$$f(\mathbf{x}) := \arg \max_{j \in [c]} \sum_{i=1}^k \phi_i \cdot \mathbb{I}[f_i(\mathbf{x}) = j] \quad (11)$$

where $\mathbb{I}$ denotes the identity function.

B.2 DISTRIBUTION FORMALIZATIONS

This subsection renders precise the distribution definitions adumbrated in the main paper.

Empirical Distributions. The distribution $\mathcal{D}$ over the input features is defined on the basis of a dataset $\mathbf{D}$ comprising diverse inputs $\mathbf{z}^1, \mathbf{z}^2, \dots, \mathbf{z}^{|\mathbf{D}|}$. Here, the probability accorded to any given input $\mathbf{x} \in \mathbb{F}$ is dictated by the frequency with which $\mathbf{x}$ occurs within $\mathbf{D}$, specifically by:

$$\Pr(\mathbf{x}) := \frac{1}{|\mathbf{D}|} \sum_{i=1}^{|\mathbf{D}|} \mathbb{I}(\mathbf{z}^i = \mathbf{x}) \quad (12)$$

Independent Distributions. Formally, given a probability value $p(\mathbf{x}_i) \in [0, 1]$ prescribed for each individual input feature, we say that the distribution $\mathcal{D}$ is independent whenever the joint probability over inputs factorizes as $\Pr(\mathbf{x}) := \prod_{i \in [n]} p(\mathbf{x}_i)$.

We observe that, when $p(\mathbf{x}_i) = p(\mathbf{x}_j)$ holds for all $i, j \in [n]$, the distribution collapses to the *uniform distribution*, itself a special case of the independent distributions.

General Distributions. We remark that many of the proofs in this work — particularly those bearing upon the fundamental properties of value functions — apply broadly across general distributional assumptions. While empirical distributions (which employ the training dataset as a proxy) are commonplace in XAI, alternative frameworks for approximating distributions encompass k-NN resampling from proximate points (Li et al., 2023; Almanjahie et al., 2018; Gweon et al., 2019), copulas for the modeling of tabular dependencies (Größer & Okhrin, 2022), and more sophisticated conditional generative models such as CTGAN (Xu et al., 2019) and conditional diffusion models (Huang et al., 2022). Choices of this nature recur throughout the counterfactual explanation and algorithmic recourse literature (Karimi et al., 2021; Fokkema et al., 2024; 2023; Verma et al., 2024), which is itself allied to contrastive explanations. Furthermore, whenever the value function is not computed directly from a structural property of the model (e.g., leaf enumeration in decision trees) or from empirical distributions, one may instead approximate it by methods such as Monte Carlo sampling (Hastings, 1970), following approaches akin to those of (Subercaseaux et al., 2025; Lopardo et al., 2023).

B.3 SUBSET VS. CARDINAL MINIMALITY

In this subsection, we furnish a more detailed discussion of the distinction between cardinal and subset minimality. Cardinal minimality confers a substantially stronger guarantee, inasmuch as it corresponds to a globally minimal explanation size, whereas subset minimality secures only a local form of minimality.

To apprehend why cardinal minimality is the stronger notion, observe that if $S \subseteq [n]$ is a cardinally minimal explanation, then no smaller set S' with $|S'| < |S|$ can qualify as an explanation; a fortiori, no strict subset $S' \subsetneq S$ is an explanation, whence S is also subset minimal.

The converse, however, does not hold. Consider the function:

$$f := \mathbf{x}_1 \vee (\mathbf{x}_2 \wedge \mathbf{x}_3) \quad (13)$$

together with the assignment $\mathbf{x} := (1, 1, 1)$, which yields $f(1, 1, 1) = 1$. Fixing the feature $\mathbf{x}_1 = 1$ constitutes a cardinally minimal (and hence subset minimal) sufficient reason, since the prediction abides at 1 irrespective of $\mathbf{x}_2, \mathbf{x}_3$. Yet fixing both $\mathbf{x}_2 = 1$ and $\mathbf{x}_3 = 1$ likewise constitutes a subset minimal explanation — though decidedly not a cardinally minimal one. Thus, while every cardinally minimal explanation is subset minimal, not every subset minimal explanation is, conversely, cardinally minimal.

C PROOF OF PROPOSITION 1

Proposition 1. *While the non-probabilistic case ($\delta = 1$) admits a unique subset-minimal global sufficient reason (Bassan et al., 2024), in the general probabilistic setting (for arbitrary δ), there exist functions f that have $\Theta(\frac{2^n}{\sqrt{n}})$ subset-minimal global sufficient reasons.*

Proof. It is known that certain Boolean functions admit an exponential number of subset-minimal local (non-probabilistic) sufficient reasons for some input $\mathbf{x}$ Bassan et al. (2024); this phenomenon was, in particular, shown to arise for functions of the following form. We demonstrate that the very same function admits an exponential number of *global* and *probabilistic* sufficient reasons. Notably, the function in question is a special case of a broader class of *threshold Boolean functions* treated in Wegener (2005). Specifically, for $n = 2k + 1$ with $k \in \mathbb{N}$, the function is defined as:

$$f(\mathbf{x}) := \begin{cases} 1 & \text{if } \sum_{i=1}^n \mathbf{x}_i \geq k + 1 \\ 0 & \text{otherwise} \end{cases} \quad (14)$$

where $\mathbf{x}$ is drawn from a uniform distribution $\mathcal{D}$. Notably, the function f is *symmetric*, in the sense of the symmetric threshold Boolean functions of Wegener (2005): its output is contingent solely upon the number of 1's (or, equivalently, 0's) in the input and remains invariant under any permutation of the input bits. Furthermore, the two extreme inputs — the all-zeros vector $(0, \dots, 0)$ and the all-ones vector $(1, \dots, 1)$ — each admit an exponential number of subset-minimal local (non-probabilistic) sufficient reasons.

Since the function f is symmetric, the condition $v_{\text{suff}}^g(S_1) = v_{\text{suff}}^g(S_2)$ obtains for any two subsets $S_1, S_2 \subseteq [n]$. Moreover, one may readily verify that, for any subset $S \subseteq [n]$ of size $k + 1$, we have:

$$(\forall i \in S). \quad v_{\text{suff}}^g(S) > v_{\text{suff}}^g(S \setminus \{i\}) \quad (15)$$

We also deliberately set δ to satisfy:

$$v_{\text{suff}}^g(S \setminus \{i\}) < \delta \leq v_{\text{suff}}^g(S), \quad (16)$$

and further set $k := \lfloor \frac{n}{2} \rfloor$. Each of these subsets is *subset-minimal*, for the reason that any subset of size at most $\lfloor \frac{n}{2} \rfloor - 1$ fails to constitute a sufficient reason for $\langle f, \mathbf{x} \rangle$ (with respect to the δ threshold). We may therefore bring to bear the identical analysis as in (Bassan et al., 2024): in particular, there exist exactly $\binom{n}{\lfloor \frac{n}{2} \rfloor}$ subset-minimal local sufficient reasons for $\langle f, \mathbf{x} \rangle$. Invoking Stirling's approximation, we obtain:

$$\lim_{n \rightarrow \infty} \frac{2\sqrt{2\pi}}{e^2} \cdot \frac{2^n}{\sqrt{n}} \leq \binom{n}{\lfloor \frac{n}{2} \rfloor} \leq \frac{e}{\pi} \cdot \frac{2^n}{\sqrt{n}} \quad (17)$$

This yields, in turn, the corresponding bound on the number of subset-minimal global δ -sufficient reasons.

□

D PROOF OF PROPOSITION 2

Proposition 2. *While the local probabilistic setting (for any δ) lacks monotonicity — i.e., the value functions v_{con}^l and v_{suff}^l are non-monotonic (Arenas et al., 2022; Izza et al., 2023; Subercaseaux et al., 2025; Izza et al., 2024b) — in the global probabilistic setting (also for any δ), both value functions v_{con}^g and v_{suff}^g are monotonic non-decreasing.*

Proof. In this section, we establish the monotonicity property for both varieties of global explanation — sufficient and contrastive alike. We commence by demonstrating the property for global sufficiency. So as to cultivate intuition, we first confine our attention to the simpler case of Boolean functions, and thereafter generalize the result to functions defined over discrete multi-valued input domains with multiple output classes. We subsequently extend the result to continuous input domains, thereby showing that the monotonicity property holds for any well-defined classification function. Finally, we exhibit how the very same monotonicity guarantees carry over to global contrastive explanations. Crucially, none of our proofs is predicated upon the assumption of feature independence — the monotonicity property holds robustly for *any* underlying distribution.

Lastly, we adduce a simple example demonstrating that the monotonicity property fails to hold for the local probabilistic sufficient or contrastive value functions.

Lemma 1. *The global sufficient value function v_{suff}^g is monotonic non-decreasing for Boolean functions, under any data distribution.*

Proof. Given an arbitrary set $S \in [n]$, and a feature $i \notin S$, we have that:

$$\Pr(\mathbf{x}) = \Pr(\mathbf{x}_S, \mathbf{x}_{\bar{S}}) = \Pr(\mathbf{x}_{\bar{S}} \mid \mathbf{x}_S) \cdot \Pr(\mathbf{x}_S) \quad (18)$$

Consequently, this implies that:

$$\begin{aligned} \Pr(\mathbf{x}) &= \Pr(\mathbf{x}_{\overline{S \cup \{i\}}} \mid \mathbf{x}_{S \cup \{i\}}) \cdot \Pr(\mathbf{x}_{S \cup \{i\}}) \\ &= \Pr(\mathbf{x}_{\overline{S \cup \{i\}}} \mid \mathbf{x}_{S \cup \{i\}}) \cdot \Pr(\mathbf{x}_S, x_i) \\ &= \Pr(\mathbf{x}_{\overline{S \cup \{i\}}} \mid \mathbf{x}_{S \cup \{i\}}) \cdot \Pr(x_i \mid \mathbf{x}_S) \cdot \Pr(\mathbf{x}_S) \end{aligned} \quad (19)$$

Moreover, for any two points $\mathbf{x}, \mathbf{x}' \sim \mathcal{D}$ such that $\mathbf{x}_S = \mathbf{x}'_S$ and $f(\mathbf{x}) = 1 - f(\mathbf{x}')$, it holds that:

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S) = 1 - \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}') \mid \mathbf{z}_S = \mathbf{x}'_S) \quad (20)$$

For the sake of notational economy, we shall on occasion suppress the distribution notation $\sim \mathcal{D}$ within the local probability expression $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)$, and shall employ f^+ and f^- to denote the events $f(\mathbf{z}) = 1$ and $f(\mathbf{z}) = 0$, respectively.

We begin with a technical simplification of the probability term, calculated to facilitate the proof, and thereafter proceed to establish monotonicity.

Simplifying the probability term. Let $\mathcal{D}_S$ denote the distribution restricted to the set S . Given a point $\mathbf{x} \sim \mathcal{D}$ and a set S , consider all points $\mathbf{z} \sim \mathcal{D}$ for which $\mathbf{z}_S = \mathbf{x}_S$ and $f(\mathbf{z}) = b$, where $b \in \{0, 1\}$ — all such points share one and the same local probability $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = b \mid \mathbf{z}_S = \mathbf{x}_S)$. For every $\mathbf{x} \sim \mathcal{D}$ sharing the same x_S and the same output b , we marginalize over $x_{\bar{S}}$. This yields $\Pr(f(\mathbf{x}) = b, \mathbf{x}_S) = \Pr(f(\mathbf{x}) = b \mid \mathbf{x}_S) \cdot \Pr(\mathbf{x}_S)$, whence we may infer:

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = b \mid \mathbf{z}_S = \mathbf{x}_S) = \Pr(f(\mathbf{x}) = b \mid \mathbf{x}_S). \quad (21)$$

Thus, for $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)]$, we get that:

$$\begin{aligned}
& \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S, f^+} \Pr(f^+, \mathbf{x}_S) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S) + \sum_{\mathbf{x}_S \sim \mathcal{D}_S, f^-} \Pr(f^-, \mathbf{x}_S) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f^- \mid \mathbf{z}_S = \mathbf{x}_S) \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [(\Pr_{\mathbf{z} \sim \mathcal{D}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S))^2 + (\Pr_{\mathbf{z} \sim \mathcal{D}}(f^- \mid \mathbf{z}_S = \mathbf{x}_S))^2] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [(\Pr_{\mathbf{z} \sim \mathcal{D}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S))^2 + (1 - \Pr_{\mathbf{z} \sim \mathcal{D}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S))^2],
\end{aligned} \tag{22}$$

From which it follows that:

$$\begin{aligned}
& \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] \\
&= \sum_{\mathbf{x}_{S \cup \{t\}} \sim \mathcal{D}_{S \cup \{t\}}} \Pr(\mathbf{x}_{S \cup \{t\}}) \cdot [(\Pr_{\mathbf{z}}(f^+ \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}}))^2 + (1 - \Pr_{\mathbf{z}}(f^+ \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}}))^2] \\
&= \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t=1}} \Pr(x_t = 1, \mathbf{x}_S) \cdot [(\Pr_{\mathbf{z}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S, z_t = 1))^2 + (1 - \Pr_{\mathbf{z}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S, z_t = 1))^2] + \\
&\quad \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t=0}} \Pr(x_t = 0, \mathbf{x}_S) \cdot [(\Pr_{\mathbf{z}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S, z_t = 0))^2 + (1 - \Pr_{\mathbf{z}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S, z_t = 0))^2]
\end{aligned} \tag{23}$$

Proving monotonicity. We now proceed to establish the monotonicity claim, building upon the foregoing simplification. We first introduce a modicum of additional notation. Let $g_1^+ := \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid \mathbf{z}_S = \mathbf{x}_S, z_t = 1)$ and $g_0^+ := \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid \mathbf{z}_S = \mathbf{x}_S, z_t = 0)$. Similarly, define $P_1 := \Pr(z_t = 1 \mid \mathbf{x}_S)$ and $P_0 := \Pr(z_t = 0 \mid \mathbf{x}_S)$. Then, the following expectations $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)]$ and $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})]$ can be simplified accordingly¹:

$$\begin{aligned}
& \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [(P_1 \cdot g_1^+ + P_0 \cdot g_0^+)^2 + (1 - (P_1 \cdot g_1^+ + P_0 \cdot g_0^+))^2] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [2(P_1 \cdot g_1^+ + P_0 \cdot g_0^+)^2 - 2(P_1 \cdot g_1^+ + P_0 \cdot g_0^+) + 1].
\end{aligned} \tag{24}$$

As a result, we obtain the following:

$$\begin{aligned}
& \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] \\
&= \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t=1}} \Pr(\mathbf{x}_S) \cdot P_1 \cdot ((g_1^+)^2 + (1 - g_1^+)^2) + \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t=0}} \Pr(\mathbf{x}_S) \cdot P_0 \cdot ((g_0^+)^2 + (1 - g_0^+)^2) \\
&= \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t=1}} \Pr(\mathbf{x}_S) \cdot P_1 \cdot (2(g_1^+)^2 - 2g_1^+ + 1) + \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t=0}} \Pr(\mathbf{x}_S) \cdot P_0 \cdot (2(g_0^+)^2 - 2g_0^+ + 1) \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [P_1 \cdot (2(g_1^+)^2 - 2g_1^+ + 1) + P_0 \cdot (2(g_0^+)^2 - 2g_0^+ + 1)] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [2P_1 \cdot ((g_1^+)^2 - g_1^+) + 2P_0 \cdot ((g_0^+)^2 - g_0^+) + 1] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [2(P_1 \cdot (g_1^+)^2 + P_0 \cdot (g_0^+)^2) - 2(P_1 \cdot g_1^+ + P_0 \cdot g_0^+) + 1].
\end{aligned} \tag{25}$$

¹Note that the notations P_1 , P_0 , g_1^+ , and g_0^+ —as well as all others used throughout the paper—are not fixed constants, but rather depend on the specific input $\mathbf{x}_S$. Formally, they should be written as $P_1(\mathbf{x}_S)$, $P_0(\mathbf{x}_S)$, and so on. However, for readability, we omit the explicit dependence on $\mathbf{x}_S$.

Let $\Delta_t(S, f)$ denote $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] - \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)]$. Without loss of generality, we assume that both P_1 and P_0 are positive. Otherwise, it can be verified that $\Delta_t(S, f) = 0$. Then, we get:

$$\begin{aligned} \Delta_t(S, f) &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \mathbf{Pr}(\mathbf{x}_S) \cdot [2(P_1 \cdot (g_1^+)^2 + P_0 \cdot (g_0^+)^2) - 2(P_1 \cdot g_1^+ + P_0 \cdot g_0^+)^2] \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \mathbf{Pr}(\mathbf{x}_S) \cdot [(P_1 - P_1^2)(g_1^+)^2 + (P_0 - P_0^2)(g_0^+)^2 - 2 \cdot P_1 \cdot P_0 \cdot g_1^+ \cdot g_0^+] \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \mathbf{Pr}(\mathbf{x}_S) \cdot P_1 \cdot P_0 \cdot (g_1^+ - g_0^+)^2 \\ &\geq 0. \end{aligned} \tag{26}$$

Observe that $P_1 - P_1^2 = P_1(1 - P_1) = P_1 P_0$ and, by the same token, $P_0 - P_0^2 = P_0(1 - P_0) = P_0 P_1$. Marshalling these identities in conjunction with the preceding result, we conclude that $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)]$ is monotone, thereby bringing our proof to a close. $\square$

Lemma 2. *The global sufficient value function v_{diff}^g is monotonic non-decreasing for functions with discrete multi-valued input domains and multiple output classes, under any data distribution.*

Proof. Extending the proof from the simplified Boolean case, we now generalize the result to functions whose input domains and output ranges each comprise a finite set of values. Let $\mathcal{K}$ denote the set of output classes.

We can generalize equation 22 as follows:

$$\begin{aligned} \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] &= \\ \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \mathbf{Pr}(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} (\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S))^2 \right] \end{aligned} \tag{27}$$

Given the condition $\mathbf{z}_S = \mathbf{x}_S$, suppose feature t can take r possible values $\{v_1, \dots, v_r\}$. Then, equation 23 can be generalized as follows:

$$\begin{aligned} \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] &= \\ = \sum_{k=1}^r \left[\sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t = v_k}} \mathbf{Pr}(\mathbf{x}_S) \cdot \mathbf{Pr}(z_t = v_k \mid \mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} (\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S, z_t = v_k))^2 \right] \right]. \end{aligned} \tag{28}$$

Let P_k denote $\mathbf{Pr}(z_t = v_k \mid \mathbf{x}_S)$, Let g_k^j denote $\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S, z_t = v_k)$, that is, z_t takes the value v_k and the output class is j . Then, we have

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] = \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \mathbf{Pr}(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \left(\sum_{k=1}^r P_k \cdot g_k^j \right)^2 \right] \tag{29}$$

and

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] = \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \mathbf{Pr}(\mathbf{x}_S) \cdot \left[\sum_{k=1}^r P_k \cdot \left(\sum_{j \in \mathcal{K}} (g_k^j)^2 \right) \right]$$

Combining these two implications, we can conclude that:

$$\begin{aligned}
\Delta_t(S, f) &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \left[\sum_{k=1}^r P_k \cdot \left(\sum_{j \in \mathcal{K}} (g_k^j)^2 \right) - \sum_{j \in \mathcal{K}} \left(\sum_{k=1}^r P_k \cdot g_k^j \right)^2 \right] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \left(\sum_{k=1}^r P_k \cdot (g_k^j)^2 \right) - \sum_{j \in \mathcal{K}} \left(\sum_{k=1}^r (P_k \cdot g_k^j)^2 + \sum_{k < l} 2 \cdot P_k \cdot P_l \cdot g_k^j \cdot g_l^j \right) \right] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \sum_{k \neq l} P_k \cdot P_l \cdot (g_k^j)^2 - \sum_{j \in \mathcal{K}} \sum_{k < l} 2 \cdot P_k \cdot P_l \cdot g_k^j \cdot g_l^j \right] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \sum_{j \in \mathcal{K}} \left[\sum_{k < l} P_k \cdot P_l \cdot (g_k^j - g_l^j)^2 \right] \\
&\geq 0.
\end{aligned} \tag{30}$$

Hence $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)]$ is monotone, which brings the proof for this setting to completion. $\square$

Lemma 3. *The global sufficient value function v_{uff}^g is monotonic non-decreasing for functions with continuous input domains and multiple output classes, under any data distribution.*

Proof. We extend the monotonicity results to functions whose features are defined over continuous domains. One avenue is to partition the domain of the continuous features into two disjoint subdomains and thereafter adapt the proofs in lemma 2 by revising the notation accordingly; the core reasoning persists unaltered in the passage from the discrete to the continuous case — only the notation is affected.

Alternatively, we furnish a direct proof below. Assume that all features reside in continuous domains, and let $p(\mathbf{x}_S, \mathbf{x}_{\bar{S}})$ denote the corresponding joint probability density function. Let $p(x_i)$, moreover, denote the marginal probability density function of x_i , and let $\mathcal{X}_i$ denote the domain of feature i .

We define marginal density as

$$p(\mathbf{x}_S) := \int_{\mathbf{x}_{\bar{S}} \in \mathcal{X}_{\bar{S}}} p(\mathbf{x}_S, \mathbf{x}_{\bar{S}}) d\mathbf{x}_{\bar{S}}, \tag{31}$$

where $\mathcal{X}_{\bar{S}}$ represents the domain of features in $\bar{S}$.

And the conditional density is defined as:

$$p(x_t \mid \mathbf{x}_S) := \frac{p(\mathbf{x}_S, x_t)}{p(\mathbf{x}_S)}. \tag{32}$$

The local probability of S given $\mathbf{x}$ is defined as

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S) = \frac{\int_{\mathbf{z}_{\bar{S}} \in \mathcal{X}_{\bar{S}}} I(f(\mathbf{x}_S, \mathbf{z}_{\bar{S}}) = j) \cdot p(\mathbf{x}_S, \mathbf{z}_{\bar{S}}) d\mathbf{z}_{\bar{S}}}{p(\mathbf{x}_S)}, \tag{33}$$

This term can also be decomposed into:

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S) = \int_{z_t \in \mathcal{X}_t} p(z_t \mid \mathbf{x}_S) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S, z_t = x_t) dz_t \tag{34}$$

via an additional feature $t \notin S$.

We then modify equation 27 to

$$\begin{aligned}
&\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\
&= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} (\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S))^2 \right] d\mathbf{x}_S,
\end{aligned} \tag{35}$$

while equation 28 is changed to

$$\begin{aligned} & \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\int_{z_t \in \mathcal{X}_t} p(z_t \mid \mathbf{x}_S) \cdot \left(\sum_{j \in \mathcal{K}} (\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}}))^2 \right) dz_t \right] d\mathbf{x}_S. \end{aligned} \quad (36)$$

For simplicity, let $Y_j(t)$ denote $\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S, z_t = v_t)$, and $p_{t|S}$ denote $p(z_t \mid \mathbf{x}_S)$. Note that both $Y_j(t)$ and $p_{t|S}$ are associated with a fixed $\mathbf{x}_S$ ². We prove monotonicity as follows:

$$\begin{aligned} & \Delta_t(S, f) \\ &= \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] - \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\int_{z_t \in \mathcal{X}_t} p_{t|S} \cdot \left(\sum_{j \in \mathcal{K}} (\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}}))^2 \right) dz_t \right] d\mathbf{x}_S - \\ & \quad \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \left(\int_{z_t \in \mathcal{X}_t} p_{t|S} \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S, z_t = x_t) dz_t \right)^2 \right] d\mathbf{x}_S \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \left(\int_{\mathcal{X}_t} p_{t|S} \cdot Y_j(t)^2 dz_t - \left(\int_{\mathcal{X}_t} p_{t|S} \cdot Y_j(t) dz_t \right)^2 \right) \right] d\mathbf{x}_S \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \frac{2}{2} \cdot \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_{t|S} \cdot p_{t'|S} \cdot (Y_j(t)^2 - Y_j(t)Y_j(t')) dz_t dz_{t'} \right) \right] d\mathbf{x}_S \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \frac{1}{2} \cdot \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_{t|S} \cdot p_{t'|S} \cdot (Y_j(t) - Y_j(t'))^2 dz_t dz_{t'} \right) \right] d\mathbf{x}_S \\ &\geq 0. \end{aligned} \quad (37)$$

□

Proposition 2. While the local probabilistic setting (for any δ) lacks monotonicity — i.e., the value functions v_{con}^ℓ and v_{suff}^ℓ are non-monotonic (Arenas et al., 2022; Izza et al., 2023; Subercaseaux et al., 2025; Izza et al., 2024b) — in the global probabilistic setting (also for any δ), both value functions v_{con}^g and v_{suff}^g are monotonic non-decreasing.

Proof. As a consequence of lemmas 1 to 3, we conclude that the global sufficient value function v_{suff}^g is monotonic non-decreasing for any well-defined classification function — whether its features be Boolean, discrete, continuous, or some combination thereof — under any data distribution whatsoever.

$$v_{con}^g(S) = 1 - v_{suff}^g(\bar{S}) \quad (38)$$

This entails that v_{con}^g is monotonic non-increasing with respect to the adjunction of features to $\bar{S}$. Equivalently, v_{con}^g is monotonic non-decreasing as features are excised from $\bar{S}$ and transferred to S , which thereby completes the proof.

□

E PROOF OF PROPOSITION 3

Proposition 3. While the local probabilistic sufficient setting (for any δ) lacks supermodularity — even when $\mathcal{D}$ is uniform, i.e., the value function v_{suff}^ℓ is not supermodular — in the global probabilistic setting (also for any δ), when $\mathcal{D}$ exhibits feature independence, the value function v_{suff}^g is supermodular.

Proof. This section establishes the supermodularity property for both global sufficient and contrastive explanations. As with monotonicity, we commence with Boolean functions and generalize by degrees

²Changing $\mathbf{x}_S$ alters their values, we omit $\mathbf{x}_S$ in the notation for brevity.

to multi-valued discrete and continuous input domains, culminating in the case of well-defined classification functions. In contradistinction to the monotonicity property, however, these proofs are contingent upon the assumption of feature independence. We further adduce a counterexample demonstrating that supermodularity may founder once this assumption is relaxed.

Lemma 4. *The global sufficient value function v_{uff}^g is supermodular for Boolean functions, under the assumption of feature independence.*

Proof. Specifically, we shall prove that:

$$\Delta_t(S, f) \leq \Delta_t(S', f), \quad (39)$$

where $S \subseteq S'$ and $t \notin S'$ — that is, that $\Delta_t(S, f)$ is monotone.

Manifestly, if $S' = S$, then $\Delta_t(S, f) \leq \Delta_t(S', f)$ holds trivially. Let $S' = S \cup \{i\}$, where $i \notin S$ and $i \neq t$. For the local probability $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)$, under the assumption of feature independence, the following holds:

$$\begin{aligned} & \Pr_{\mathbf{z}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S) \\ &= \Pr(z_i = 1) \cdot \Pr(z_t = 1) \cdot \Pr_{\mathbf{z}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 1, z_t = 1) + \\ & \quad \Pr(z_i = 1) \cdot \Pr(z_t = 0) \cdot \Pr_{\mathbf{z}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 1, z_t = 0) + \\ & \quad \Pr(z_i = 0) \cdot \Pr(z_t = 1) \cdot \Pr_{\mathbf{z}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 0, z_t = 1) + \\ & \quad \Pr(z_i = 0) \cdot \Pr(z_t = 0) \cdot \Pr_{\mathbf{z}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 0, z_t = 0). \end{aligned} \quad (40)$$

Let P_1, P_0, Q_1 , and Q_0 denote $\Pr(z_t = 1), \Pr(z_t = 0), \Pr(z_i = 1)$, and $\Pr(z_i = 0)$, respectively. Moreover, let $g_{11}^+, g_{10}^+, g_{01}^+$, and g_{00}^+ denote $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 1, z_t = 1), \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 1, z_t = 0), \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 0, z_t = 1)$, and $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 0, z_t = 0)$, respectively.

We further decompose equation 26 into:

$$\begin{aligned} \Delta_t(S, f) &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \Pr(\mathbf{x}_S) \cdot P_1 P_0 \cdot [(Q_1 \cdot g_{11}^+ + Q_0 \cdot g_{01}^+) - (Q_1 \cdot g_{10}^+ + Q_0 \cdot g_{00}^+)]^2 \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \Pr(\mathbf{x}_S) \cdot P_1 P_0 \cdot [Q_1 \cdot (g_{11}^+ - g_{10}^+) + Q_0 \cdot (g_{01}^+ - g_{00}^+)]^2. \end{aligned} \quad (41)$$

Similarly, the following also holds:

$$\Delta_t(S \cup \{i\}, f) = \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \Pr(\mathbf{x}_S) \cdot P_1 P_0 \cdot [Q_1 \cdot (g_{11}^+ - g_{10}^+)^2 + Q_0 \cdot (g_{01}^+ - g_{00}^+)^2]. \quad (42)$$

Thus, we get

$$\begin{aligned} \Delta_t(S \cup \{i\}, f) - \Delta_t(S, f) &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \Pr(\mathbf{x}_S) \cdot P_1 P_0 \cdot Q_1 Q_0 \cdot (g_{11}^+ - g_{10}^+ - g_{01}^+ + g_{00}^+)^2 \\ &\geq 0. \end{aligned} \quad (43)$$

This entails that $\Delta_t(S, f)$ is monotone — that is, $c(S \cup \{t\}, f) - c(S, f) \leq c(S' \cup \{t\}, f) - c(S', f)$. $\square$

Lemma 5. *The global sufficient value function v_{uff}^g is supermodular for functions with discrete multi-valued input domains and multiple output classes, under the assumption of feature independence.*

Proof. We extend the results of the foregoing section to multi-valued discrete functions, once again under the assumption of feature independence. We use P_k to denote $\Pr(z_t = v_k)$, and $Q_{k'}$ to denote

$\Pr(z_i = v'_k)$. Moreover, we use $g_{k'k}^j$ to denote $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S, z_i = v'_k, z_t = v_k)$. We decompose equation 30 to

$$\begin{aligned} \Delta_t(S, f) &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \sum_{j \in \mathcal{K}} \left[\sum_{k < l} P_k P_l \cdot \left(\sum_{k'=1}^{r'} Q_{k'} \cdot g_{k'k}^j - \sum_{k'=1}^{r'} Q_{k'} \cdot g_{k'l}^j \right)^2 \right] \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \sum_{j \in \mathcal{K}} \left[\sum_{k < l} P_k P_l \cdot \left(\sum_{k'=1}^{r'} Q_{k'} \cdot (g_{k'k}^j - g_{k'l}^j) \right)^2 \right]. \end{aligned} \quad (44)$$

Likewise, we have

$$\begin{aligned} \Delta_t(S \cup \{i\}, f) &= \sum_{k'=1}^{r'} \left[\sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_i = v'_k}} \Pr(\mathbf{x}_S) \cdot \Pr(z_i = v'_k) \cdot \sum_{j \in \mathcal{K}} \left(\sum_{k < l} P_k P_l \cdot (g_{k'k}^j - g_{k'l}^j)^2 \right) \right] \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \left[\sum_{k'=1}^{r'} Q_{k'} \cdot \sum_{j \in \mathcal{K}} \left(\sum_{k < l} P_k P_l \cdot (g_{k'k}^j - g_{k'l}^j)^2 \right) \right] \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \sum_{j \in \mathcal{K}} \left[\sum_{k < l} P_k P_l \cdot \sum_{k'=1}^{r'} Q_{k'} \cdot (g_{k'k}^j - g_{k'l}^j)^2 \right]. \end{aligned} \quad (45)$$

Therefore, we get

$$\begin{aligned} \Delta_t(S \cup \{i\}, f) - \Delta_t(S, f) &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \sum_{j \in \mathcal{K}} \left[\sum_{k < l} \sum_{k' < l'} P_k P_l Q_{k'} Q_{l'} \cdot (g_{k'k}^j - g_{k'l}^j - g_{l'k}^j + g_{l'l}^j)^2 \right] \\ &\geq 0. \end{aligned} \quad (46)$$

□

Proposition 3. *While the local probabilistic sufficient setting (for any δ) lacks supermodularity — even when $\mathcal{D}$ is uniform, i.e., the value function v_{diff}^δ is not supermodular — in the global probabilistic setting (also for any δ), when $\mathcal{D}$ exhibits feature independence, the value function v_{diff}^g is supermodular.*

Proof. We may generalize the supermodularity results established in Lemma 5 to the continuous case, on the assumption that $\mathcal{D}_p$ satisfies feature independence. One avenue is to partition the domain of the continuous features into two disjoint subdomains and thereafter adapt the proofs in Lemma 5 by revising the notation accordingly.

Alternatively, we furnish a direct proof below. Under this assumption, the marginal and conditional densities admit the following simplification:

$$p(\mathbf{x}_S) = \prod_{i \in S} p(x_i), \text{ and } p(x_t \mid \mathbf{x}_S) = p(x_t). \quad (47)$$

Consequently, the local probability of S conditioned on $\mathbf{x}$ becomes:

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S) = \int_{\mathbf{z}_{\bar{S}} \in \mathcal{X}_{\bar{S}}} I(f(\mathbf{x}_S, \mathbf{z}_{\bar{S}}) = j) \cdot \prod_{i \in \bar{S}} p(z_i) d\mathbf{z}_{\bar{S}}. \quad (48)$$

We let $Y_j(i, t)$ denote $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S, z_i = x_i, z_t = x_t)$, where $i \notin S, t \notin S$ and $i \neq t$. Evidently,

$$Y_j(i, t) = Y_j(t, i), \text{ and } Y_j(t) = \int_{\mathcal{X}_i} p_i \cdot Y_j(i, t) dz_i.$$

We have:

$$\begin{aligned}
\Delta_t(S, f) &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \frac{1}{2} \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} (Y_j(t) - Y_j(t'))^2 dz_t dz_{t'} \right) \right] d\mathbf{x}_S \\
&= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \frac{1}{2} \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} \left(\int_{\mathcal{X}_i} p_i \cdot Y_j(i, t) dz_i - \int_{\mathcal{X}_i} p_i \cdot Y_j(i, t') dz_i \right)^2 dz_t dz_{t'} \right) \right] d\mathbf{x}_S \\
&= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \frac{1}{2} \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} \left(\int_{\mathcal{X}_i} p_i \cdot (Y_j(i, t) - Y_j(i, t')) dz_i \right)^2 dz_t dz_{t'} \right) \right] d\mathbf{x}_S,
\end{aligned} \tag{49}$$

where

$$\begin{aligned}
&\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} \left(\int_{\mathcal{X}_i} p_i \cdot (Y_j(i, t) - Y_j(i, t')) dz_i \right)^2 dz_t dz_{t'} \\
&= \int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} \int_{\mathcal{X}_i} \int_{\mathcal{X}_{i'}} p_t p_{t'} p_i p_{i'} \cdot (Y_j(i, t) - Y_j(i, t')) \cdot (Y_j(i', t) - Y_j(i', t')) dz_i dz_{i'} dz_t dz_{t'}.
\end{aligned}$$

Let $S' = S \cup \{i\}$, then we have:

$$\begin{aligned}
\Delta_t(S', f) &= \int_{\mathbf{x}_{S'} \in \mathcal{X}_{S'}} p(\mathbf{x}_{S'}) \cdot \left[\sum_{j \in \mathcal{K}} \frac{1}{2} \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} \cdot (Y_j(i, t) - Y_j(i, t'))^2 dz_t dz_{t'} \right) \right] d\mathbf{x}_{S'} \\
&= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \frac{1}{2} \int_{\mathcal{X}_i} p_i \cdot \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} \cdot (Y_j(i, t) - Y_j(i, t'))^2 dz_t dz_{t'} \right) dz_i \right] d\mathbf{x}_S \tag{50} \\
&= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \frac{1}{2} \left(\int_{\mathcal{X}_i} \int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_i p_t p_{t'} \cdot (Y_j(i, t) - Y_j(i, t'))^2 dz_t dz_{t'} dz_i \right) \right] d\mathbf{x}_S,
\end{aligned}$$

where

$$\begin{aligned}
&\int_{\mathcal{X}_i} \int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_i p_t p_{t'} \cdot (Y_j(i, t) - Y_j(i, t'))^2 dz_t dz_{t'} dz_i \\
&= \int_{\mathcal{X}_i} \int_{\mathcal{X}_{i'}} \int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_i p_{i'} p_t p_{t'} \cdot (Y_j(i, t) - Y_j(i, t'))^2 dz_t dz_{t'} dz_i dz_{i'}.
\end{aligned}$$

Let $\alpha(i, t, t')$ denote $Y_j(i, t) - Y_j(i, t')$ and $\beta(i', t, t')$ denote $Y_j(i', t) - Y_j(i', t')$, we can infer that

$$\begin{aligned}
\Delta_t(S', f) - \Delta_t(S, f) &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \left[\sum_{j \in \mathcal{K}} \frac{1}{4} \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} \int_{\mathcal{X}_i} \int_{\mathcal{X}_{i'}} p_t p_{t'} p_i p_{i'} (\alpha(i, t, t') - \beta(i', t, t'))^2 dz_i dz_{i'} dz_t dz_{t'} \right) \right] d\mathbf{x}_S \\
&\geq 0.
\end{aligned} \tag{51}$$

Consequently, the global sufficient value function v_{suff}^g retains the supermodularity property for any valid classification function, irrespective of whether its features be Boolean, discrete, continuous, or some admixture thereof.

We now demonstrate that the local probabilistic sufficient value function is neither submodular nor supermodular. By way of a simple illustration, fix the uniform distribution over the three variables $\{x_1, x_2, x_3\}$ and define a Boolean function as follows:

$$f(x_1, x_2) = (x_1 \vee x_2) \wedge x_3. \tag{52}$$

Consider the input $\mathbf{x} = (1, 1, 1)$, which is classified as 1. For the local probabilistic sufficient value function v_{suff}^l , the following holds:

$$v_{\text{suff}}^l(\emptyset) = \frac{3}{8}, \quad v_{\text{suff}}^l(\{1\}) = \frac{1}{2}, \quad v_{\text{suff}}^l(\{2\}) = \frac{1}{2}, \quad v_{\text{suff}}^l(\{3\}) = \frac{3}{4},$$

$$v_{\text{suff}}^l(\{1, 2\}) = \frac{1}{2}, \quad v_{\text{suff}}^l(\{1, 3\}) = 1, \quad v_{\text{suff}}^l(\{2, 3\}) = 1, \quad v_{\text{suff}}^l(\{1, 2, 3\}) = 1.$$

Since $(v_{\text{suff}}^l(\{1, 2\}) - v_{\text{suff}}^l(\{2\})) < (v_{\text{suff}}^l(\{1\}) - v_{\text{suff}}^l(\emptyset)) < (v_{\text{suff}}^l(\{1, 3\}) - v_{\text{suff}}^l(\{3\}))$, the function v_{suff}^l is neither supermodular nor submodular. $\square$

F PROOF OF PROPOSITION 4

Proposition 4. *While the local probabilistic contrastive setting (for any δ) lacks submodularity — even when $\mathcal{D}$ is uniform, i.e., the value function v_{con}^l is not submodular — in the global probabilistic setting (also for any δ), when $\mathcal{D}$ exhibits feature independence, the value function v_{con}^g is submodular.*

Proof. Once the global sufficient setting has been established to be supermodular, it follows immediately that the global contrastive setting is submodular. Let $S \subseteq [n]$ be the set of features permitted to change, and suppose $S \subseteq S'$ with $i \notin S'$. In order to juxtapose the marginal contributions $v_{\text{con}}^g(S \cup \{i\}) - v_{\text{con}}^g(S)$ and $v_{\text{con}}^g(S' \cup \{i\}) - v_{\text{con}}^g(S')$, we equivalently juxtapose $(1 - v_{\text{suff}}^g(\bar{S} \setminus \{i\})) - (1 - v_{\text{suff}}^g(\bar{S}))$ and $(1 - v_{\text{suff}}^g(\bar{S}' \setminus \{i\})) - (1 - v_{\text{suff}}^g(\bar{S}'))$. Since $S \subseteq S'$, it follows that $\bar{S}' \subseteq \bar{S}$, and we thereby obtain:

$$(v_{\text{suff}}^g(\bar{S}) - v_{\text{suff}}^g(\bar{S} \setminus \{i\})) \geq (v_{\text{suff}}^g(\bar{S}') - v_{\text{suff}}^g(\bar{S}' \setminus \{i\})) \quad (53)$$

Which is equivalent to:

$$(v_{\text{con}}^g(S \cup \{i\}) - v_{\text{con}}^g(S)) \geq (v_{\text{con}}^g(S' \cup \{i\}) - v_{\text{con}}^g(S')). \quad (54)$$

Thus the global contrastive value function v_{con}^g is submodular, which brings this proof to its conclusion.

We now demonstrate that the local probabilistic contrastive value function is neither submodular nor supermodular. By way of a simple illustration, fix the uniform distribution over the three variables $\{x_1, x_2, x_3\}$ and define a Boolean function as follows:

$$f(x_1, x_2) := (x_1 \vee x_2) \wedge x_3. \quad (55)$$

Consider the input $\mathbf{x} = (1, 1, 1)$, which is classified as 1. For the local probabilistic contrastive value function v_{con}^l , we have:

$$v_{\text{con}}^l(\emptyset) = 1, \quad v_{\text{con}}^l(\{1\}) = 1, \quad v_{\text{con}}^l(\{2\}) = 1, \quad v_{\text{con}}^l(\{3\}) = \frac{1}{2},$$

$$v_{\text{con}}^l(\{1, 2\}) = \frac{3}{4}, \quad v_{\text{con}}^l(\{1, 3\}) = \frac{1}{2}, \quad v_{\text{con}}^l(\{2, 3\}) = \frac{1}{2}, \quad v_{\text{con}}^l(\{1, 2, 3\}) = \frac{3}{8}.$$

Since $(v_{\text{con}}^l(\{1, 2\}) - v_{\text{con}}^l(\{2\})) < (v_{\text{con}}^l(\{1, 2, 3\}) - v_{\text{con}}^l(\{2, 3\})) < (v_{\text{con}}^l(\{1\}) - v_{\text{con}}^l(\emptyset))$, the function v_{con}^l is neither submodular nor supermodular. $\square$

In the remainder of this section, we show that the feature independence assumption is not merely sufficient but indeed necessary for supermodularity. We substantiate this with the following counterexample.

Define a Boolean function:

$$f(x_1, x_2) := x_1 \quad (56)$$

For this Boolean function, it holds that $f(00) = 0$, $f(01) = 0$, $f(10) = 1$, and $f(11) = 1$. Define, in addition, a distribution $\mathcal{D}$ over x_1 and x_2 . Recall that the global sufficient value function is given by:

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] = \sum_{\mathbf{x} \sim \mathcal{D}} \Pr(\mathbf{x}) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)$$

In addition, if we can show that

$$\Delta_t(S, f) \leq \Delta_t(S', f),$$

where $S \subseteq S'$ and $t \notin S'$, then the global sufficient value function is supermodular. When $S = \emptyset$, the following holds:

$$\begin{aligned} \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\ = \Pr(x_1 = 0, x_2 = 0) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0) + \Pr(x_1 = 0, x_2 = 1) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0) + \\ \Pr(x_1 = 1, x_2 = 0) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1) + \Pr(x_1 = 1, x_2 = 1) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1) \end{aligned}$$

Which is also equivalent to:

$$\begin{aligned} \Pr(x_1 = 0, x_2 = 0) \cdot (\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1)) + \\ \Pr(x_1 = 0, x_2 = 1) \cdot (\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1)) + \\ \Pr(x_1 = 1, x_2 = 0) \cdot (\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1)) + \\ \Pr(x_1 = 1, x_2 = 1) \cdot (\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1)) \\ = (\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1))^2 + (\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1))^2 \end{aligned}$$

When $S = \{2\}$, we have that the following condition holds:

$$\begin{aligned} \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\ = \Pr(x_1 = 0, x_2 = 0) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_2 = 0) + \\ \Pr(x_1 = 0, x_2 = 1) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_2 = 1) + \\ \Pr(x_1 = 1, x_2 = 0) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_2 = 0) + \\ \Pr(x_1 = 1, x_2 = 1) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_2 = 1) \end{aligned}$$

This can also be rearranged as follows:

$$\begin{aligned} \Pr(x_1 = 0, x_2 = 0) \cdot \frac{\Pr(x_1 = 0, x_2 = 0)}{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 1, x_2 = 0)} + \\ \Pr(x_1 = 0, x_2 = 1) \cdot \frac{\Pr(x_1 = 0, x_2 = 1)}{\Pr(x_1 = 0, x_2 = 1) + \Pr(x_1 = 1, x_2 = 1)} + \\ \Pr(x_1 = 1, x_2 = 0) \cdot \frac{\Pr(x_1 = 1, x_2 = 0)}{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 1, x_2 = 0)} + \\ \Pr(x_1 = 1, x_2 = 1) \cdot \frac{\Pr(x_1 = 1, x_2 = 1)}{\Pr(x_1 = 0, x_2 = 1) + \Pr(x_1 = 1, x_2 = 1)} \\ = \frac{\Pr(x_1 = 0, x_2 = 0)^2 + \Pr(x_1 = 1, x_2 = 0)^2}{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 1, x_2 = 0)} + \frac{\Pr(x_1 = 0, x_2 = 1)^2 + \Pr(x_1 = 1, x_2 = 1)^2}{\Pr(x_1 = 0, x_2 = 1) + \Pr(x_1 = 1, x_2 = 1)} \end{aligned}$$

In contrast, when $S = \{1\}$, we have the the following holds:

$$\begin{aligned} \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\ = \Pr(x_1 = 0, x_2 = 0) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_1 = 0) + \\ \Pr(x_1 = 0, x_2 = 1) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_1 = 0) + \\ \Pr(x_1 = 1, x_2 = 0) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_1 = 1) + \\ \Pr(x_1 = 1, x_2 = 1) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_1 = 1) \end{aligned}$$

This can similarly be rearranged as follows:

$$\begin{aligned} & \Pr(x_1 = 0, x_2 = 0) \cdot \frac{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1)}{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1)} + \\ & \Pr(x_1 = 0, x_2 = 1) \cdot \frac{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1)}{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1)} + \\ & \Pr(x_1 = 1, x_2 = 0) \cdot \frac{\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1)}{\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1)} + \\ & \Pr(x_1 = 1, x_2 = 1) \cdot \frac{\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1)}{\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1)} = 1 \end{aligned}$$

Finally, when $S = \{1, 2\}$, we have the following condition:

$$\begin{aligned} & \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\ & = \Pr(x_1 = 0, x_2 = 0) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_1 = 0, x_2 = 0) + \\ & \Pr(x_1 = 0, x_2 = 1) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_1 = 0, x_2 = 1) + \\ & \Pr(x_1 = 1, x_2 = 0) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_1 = 1, x_2 = 0) + \\ & \Pr(x_1 = 1, x_2 = 1) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_1 = 1, x_2 = 1) \end{aligned}$$

This can likewise be reordered as follows:

$$\begin{aligned} & \Pr(x_1 = 0, x_2 = 0) \cdot \frac{\Pr(x_1 = 0, x_2 = 0)}{\Pr(x_1 = 0, x_2 = 0)} + \Pr(x_1 = 0, x_2 = 1) \cdot \frac{\Pr(x_1 = 0, x_2 = 1)}{\Pr(x_1 = 0, x_2 = 1)} + \\ & \Pr(x_1 = 1, x_2 = 0) \cdot \frac{\Pr(x_1 = 1, x_2 = 0)}{\Pr(x_1 = 1, x_2 = 0)} + \Pr(x_1 = 1, x_2 = 1) \cdot \frac{\Pr(x_1 = 1, x_2 = 1)}{\Pr(x_1 = 1, x_2 = 1)} \\ & = 1 \end{aligned}$$

Let $t = 2$. Evidently, $\Delta_t(\{1\}, f) = 0$. We turn next to an analysis of $\Delta_t(\emptyset, f)$. We remark that, on the assumption of feature independence, we observe that:

$$\begin{aligned} & (\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1))^2 + (\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1))^2 \\ & = \Pr(x_1 = 0)^2 + \Pr(x_1 = 1)^2, \end{aligned}$$

Additionally, another condition that holds is:

$$\begin{aligned} & \frac{\Pr(x_1 = 0, x_2 = 0)^2 + \Pr(x_1 = 1, x_2 = 0)^2}{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 1, x_2 = 0)} + \frac{\Pr(x_1 = 0, x_2 = 1)^2 + \Pr(x_1 = 1, x_2 = 1)^2}{\Pr(x_1 = 0, x_2 = 1) + \Pr(x_1 = 1, x_2 = 1)} \\ & = \Pr(x_1 = 0)^2 + \Pr(x_1 = 1)^2. \end{aligned}$$

This entails $\Delta_t(\emptyset, f) = 0$ — that is, $\Delta_t(\emptyset, f) \leq \Delta_t(\{1\}, f)$. In the absence of the feature independence assumption, however, we cannot guarantee $\Delta_t(\emptyset, f) = 0$. By way of example, let:

$$\Pr(x_1 = 0, x_2 = 0) = P_{00}, \Pr(x_1 = 0, x_2 = 1) = P_{01}$$

and

$$\Pr(x_1 = 1, x_2 = 0) = P_{10}, \Pr(x_1 = 1, x_2 = 1) = P_{11}$$

Then, by simplifying $\Delta_t(\emptyset, f)$ we get that the following holds:

$$\Delta_t(\emptyset, f) = \left[\frac{P_{00}^2 + P_{10}^2}{P_{00} + P_{10}} + \frac{P_{01}^2 + P_{11}^2}{P_{01} + P_{11}} \right] - \left[(P_{00} + P_{01})^2 + (P_{10} + P_{11})^2 \right] \quad (57)$$

We then let $P_{00} = 0.4$, $P_{01} = 0.2$, $P_{10} = 0.1$, and $P_{11} = 0.3$, and get that the following holds:

$$\Delta_t(\emptyset, f) = \left[\frac{(0.4)^2 + (0.1)^2}{0.5} + \frac{(0.2)^2 + (0.3)^2}{0.5} \right] - \left[(0.4 + 0.2)^2 + (0.1 + 0.3)^2 \right] = 0.08. \quad (58)$$

This entails that $\Delta_t(\emptyset, f) > \Delta_t(1, f)$, which attests that — in contradistinction to the monotonicity setting, where it was dispensable — feature independence constitutes a necessary condition for supermodularity.

G PROOF OF PROPOSITION 5

Proposition 5. *Computing Algorithm 1 with the local value functions v_{con}^ℓ and v_{suff}^ℓ does not always converge to a subset minimal δ -sufficient/contrastive reason. However, computing it with the global value functions v_{con}^g or v_{suff}^g necessarily produces subset minimal δ -sufficient/contrastive reasons.*

Proof. We remark that this proposition follows immediately from our proof that both v_{suff}^g and v_{con}^g are monotone non-decreasing (as established in Proposition 2), in concert with other pertinent conclusions drawn in prior work. In the remainder of this section, we expound upon the manner in which this monotonicity property gives rise to the stated corollary.

The convergence of Algorithm 1 to a subset-minimal δ global sufficient or contrastive reason with respect to the value functions v_{con}^g and v_{suff}^g follows directly from the monotonicity property, in relation to which prior work (Ignatiev et al., 2019; Wu et al., 2024b; Arenas et al., 2022) demonstrated that a greedy algorithm of this kind yields a subset-minimal explanation whenever the underlying value function is monotone non-decreasing. The algorithm halts when, for every $i \in S$, the value $v(S \setminus \{i\})$ falls below δ , thereby ensuring that, although $v(S) \geq \delta$ (a maintained invariant), the removal of any element occasions the failure of the condition. By virtue of monotonicity, this entails that no proper subset of S satisfies the condition, whence S is guaranteed to be subset-minimal.

We now turn to elucidate why Algorithm 1 fails to converge to a subset-minimal solution in the local setting. It is well established that the local probabilistic sufficient value function (and, likewise, the local probabilistic contrastive value function) is not monotone Arenas et al. (2022); Izza et al. (2023); Subercaseaux et al. (2025); Izza et al. (2024b). By way of a simple illustration, fix the uniform distribution over the two binary variables $\{x_1, x_2\}$ and define a Boolean function as follows:

$$f(x_1, x_2) := x_1 \vee x_2. \quad (59)$$

Consider the input $\mathbf{x} = (0, 1)$, which is classified as 1; then:

$$v_{\text{suff}}^\ell(\emptyset) = \frac{3}{4}, \quad v_{\text{suff}}^\ell(\{1\}) = \frac{1}{2}, \quad v_{\text{suff}}^\ell(\{2\}) = 1, \quad v_{\text{suff}}^\ell(\{1, 2\}) = 1. \quad (60)$$

Since $v_{\text{suff}}^\ell(\emptyset) > v_{\text{suff}}^\ell(\{1\}) < v_{\text{suff}}^\ell(\{1, 2\})$, the function v_{suff}^ℓ is not monotone. By the same token, one may compute that:

$$v_{\text{con}}^\ell(\emptyset) = 1, \quad v_{\text{con}}^\ell(\{1\}) = 1, \quad v_{\text{con}}^\ell(\{2\}) = \frac{1}{2}, \quad v_{\text{con}}^\ell(\{1, 2\}) = \frac{3}{4}. \quad (61)$$

(For v_{con}^ℓ , S denotes the set of features permitted to vary.) Since $v_{\text{con}}^\ell(\emptyset) > v_{\text{con}}^\ell(\{2\}) < v_{\text{con}}^\ell(\{1, 2\})$, the function v_{con}^ℓ is likewise non-monotone.

□

H PROOF OF THEOREM 1

Theorem 1. *If f is a decision tree and the probability term v_{suff}^g can be computed in polynomial time given the distribution $\mathcal{D}$ (which holds for independent distributions, among others), then obtaining a subset-minimal global δ -sufficient reason can be obtained in polynomial time. However, unless $\text{PTIME}=\text{NP}$, no polynomial-time algorithm exists for computing a local δ -sufficient reason for decision trees even under independent distributions.*

Proof. We begin by noting that for decision trees, assuming that $\mathcal{D}$ represents independent distributions, the computation of the value function:

$$v_{\text{suff}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)] \quad (62)$$

may be carried out in polynomial time by means of the following procedure. By virtue of the tractability of decision trees, we may iterate over all pairs of leaf nodes, each corresponding to the partial assignments $\mathbf{x}_S$ and $\mathbf{z}_S$. For each such pair, in the event that both leaf nodes issue the same prediction under f , we compute the corresponding term in the expected value $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)]$ by multiplying the respective feature-wise probabilities over the shared features of the two vectors. Under the feature independence assumption, these probabilities decompose, whereby the full expectation may be computed by summing the contributions of all such matching leaf pairs.

Inasmuch as every probability involved is furnished as part of the input and each step entails only polynomial-time operations, the entire procedure runs in polynomial time. This establishes that every step of Algorithm 1 is efficient, and hence that the algorithm in its entirety runs in polynomial time. Finally, combining this with Lemma 2, which proves that v_{suff}^g is monotone non-decreasing, we conclude that Algorithm 1 converges to a subset-minimal explanation — for the algorithm halts when, for every $i \in S$, the value $v(S \setminus \{i\})$ falls below δ , while $v(S) \geq \delta$ is preserved throughout. By monotonicity, this guarantees that no strict subset of S satisfies the condition, whence S is indeed subset-minimal.

As regards the remaining part of the claim, the result follows from (Arenas et al., 2022), wherein it was shown that, on the assumption that $P \neq NP$, no polynomial-time algorithm exists for computing a subset-minimal δ -sufficient reason for decision trees. Although this was proven specifically under the uniform distribution, the hardness manifestly extends to independent distributions, which subsume this case. Taken together, these results establish the complexity separation between the local and global variants, thereby concluding our proof.

□

I PROOF OF PROPOSITION 6

Proposition 6. *For any model f , and empirical distribution $\mathcal{D}$ — computing a subset-minimal global δ -sufficient or δ -contrastive reason for f can be done in polynomial time.*

Proof. The computation of the global sufficient probability function:

$$v_{\text{suff}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)] \quad (63)$$

or the computation of the global contrastive probability function

$$v_{\text{con}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_{\bar{S}} = \mathbf{z}_{\bar{S}})] \quad (64)$$

may be performed in polynomial time whenever $\mathcal{D}$ is drawn from the class of empirical distributions. This is accomplished by iterating over pairs of instances $\mathbf{x}, \mathbf{z}$ within the dataset and running inference through $f(\mathbf{x})$ and $f(\mathbf{z})$ so as to compute the expected values at each step. Consequently, adjudicating whether the probability function exceeds or falls below a given threshold δ is likewise accomplished in polynomial time. Furthermore, drawing upon the proof of non-decreasing monotonicity (Proposition D) of the global probability function, we may deploy Algorithm 1 for either contrastive reason search or sufficient reason search. Through the execution of a linear number of polynomial queries, a subset-minimal explanation — be it a subset minimal contrastive reason or a subset minimal sufficient reason — may thus be obtained in polynomial time.

□

J PROOF OF THEOREM 2

Theorem 2. *Assuming f is a neural network or a tree ensemble, and $\mathcal{D}$ is an empirical distribution — there exist polynomial-time algorithms for obtaining subset minimal global δ -sufficient and con-*

trastive reasons. However, unless $PTIME=NP$, there is no polynomial time algorithm for computing a subset minimal local δ -sufficient reason or a subset minimal local δ -contrastive reason.

Proof. The first part of the proof follows immediately from Proposition 6, which established that this holds for any model, inasmuch as both $v_{\text{suff}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)]$ and $v_{\text{con}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_{\bar{S}} = \mathbf{z}_{\bar{S}})]$ are computable in polynomial time from the empirical distribution. As regards the second part, we proceed by establishing two lemmas: the one demonstrating the intractability of the sufficient case, the other addressing the contrastive case.

Lemma 6. *Unless $PTIME=NP$, then there is no polynomial time algorithm for computing a subset minimal local δ -sufficient reason for either a tree ensemble or a neural network, under empirical distributions.*

Proof. We shall first establish this claim for neural networks and thereafter extend the result to tree ensembles. Our argument proceeds by demonstrating that, were a polynomial-time algorithm to exist for computing a subset-minimal local δ sufficient reason for neural networks, it would thereby enable us to solve the classical NP-complete *CNF-SAT* problem in polynomial time. The *CNF-SAT* problem is defined as follows:

CNF-SAT:

Input: A formula in conjunctive normal form (CNF): ϕ .

Output: Yes, if there exists an assignment to the n literals of ϕ such that ϕ is evaluated to True, and No otherwise

Our proof will additionally avail itself of the following lemma (whose proof is provided in Barceló et al. (2020)):

Lemma 7. *Any boolean circuit ϕ can be encoded into an equivalent MLP over the binary domain $\{0, 1\}^n \rightarrow \{0, 1\}$ in polynomial time.*

Proof. We shall in fact establish hardness for a simpler, more specific case, which will consequently imply hardness for the more general setting. In this case, we assume that the empirical distribution $\mathcal{D}$ comprises a single element, which, for the sake of simplicity, we take to be the zero vector $\mathbf{0}_n$. Since the sufficiency of S in this scenario is contingent solely upon the local instance $\mathbf{x}$ and the single instance $\mathbf{0}_n$ within our empirical distribution, the nature of the input format (whether discrete, continuous, and so forth) exerts no bearing upon the result; the hardness results therefore hold universally across all these settings.

In analogy with $\mathbf{0}_n$, we define $\mathbf{1}_n$ to be an n -dimensional vector consisting entirely of ones. Given an input ϕ , we first assign $\mathbf{1}_n$ to ϕ , thereby setting all variables to True. Should ϕ evaluate to True under this assignment, a satisfying truth assignment exists; we may therefore assume that $\phi(\mathbf{1}_n) = 0$. We now introduce the following definitions:

$$\begin{aligned}\phi_2 &:= (x_1 \wedge x_2 \wedge \dots \wedge x_n), \\ \phi' &:= \phi \vee \phi_2\end{aligned}\tag{65}$$

Although ϕ' is no longer a CNF, it may nonetheless be transformed into an MLP f by means of Lemma 8, in such a manner that f behaves equivalently to ϕ' over the domain $\{0, 1\}^n$. On the strength of our assumption that computing a subset-minimal local δ -sufficient reason is feasible in polynomial time, we may determine one for the instance $\langle f, \mathbf{x} := \mathbf{1}_n \rangle$, $\delta := 1$, bearing in mind that the empirical distribution $\mathcal{D}$ is simply defined over the single data point $\mathbf{0}_n$.

We now contend that the subset-minimal δ -sufficient reason generated for $\langle f, \mathbf{x} \rangle$ encompasses the entire input space — that is, $S = \{1, \dots, n\}$ — if and only if $\langle \phi \rangle \notin \text{CNF-SAT}$.

Suppose that $S = \{1, \dots, n\}$. Since S is a δ -sufficient reason for $\langle f, \mathbf{x} \rangle$, this merely signifies that setting the complementary set to any value preserves the prediction — which, the complementary set being $\emptyset$ in this case, holds trivially. That S is subset-minimal means, in turn, that any other subset $S' \subsetneq S$ satisfies $v(S') < \delta = 1$. Since the probability function $\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)$ is determined by a single point (the distribution containing only the point $\mathbf{0}_n$), the probability function can assume the values 1 or 0 alone; hence we further know that $v(S') = 0$. This apprises us that, the

subset $S = \{1, \dots, n\}$ aside, for any subset $S' \subseteq S$, fixing the features in S' to 1 and the remainder to 0 does not yield a classification outcome of 1. Since the ϕ_2 component within ϕ' is True only if all features are assigned 1, this directly entails that ϕ is assigned False for any such input. Given that ϕ is already known not to return a True answer for the vector assignment $\mathbf{1}_n$ (as verified at the outset), and that the same has now been established for all other input vectors, we conclude that $\phi \notin \text{CNF-SAT}$.

Now suppose, conversely, that S is a subset strictly contained within $\{1, \dots, n\}$. Given that S is sufficient under our definitions of the distribution $\mathcal{D}$ with $\delta = 1$, we may invoke the same reasoning as before to conclude that $v(S) = 1$. This entails that setting the features in S to 1 while setting the remaining features to 0 ensures that the function f evaluates to 1. Since this assignment is necessarily not a vector consisting entirely of ones, it follows that the ϕ_2 component within ϕ' must be False; consequently, the ϕ component must be True, whence $\langle \phi \rangle \in \text{CNF-SAT}$. This completes the proof.

We have thus established that the claim holds for neural networks. Extending the proof to tree ensembles, however, demands only a minor and straightforward adaptation, to which end we invoke the following lemma, noted in several prior works (Ordyniak et al., 2024; Audemard et al., 2022b):

Lemma 8. *Any CNF or DNF ϕ can be encoded into an equivalent random forest classifier over the binary domain $\{0, 1\}^n \rightarrow \{0, 1\}$ in polynomial time.*

Proof. We observe that we may apply the same procedure employed in our proof for neural networks, wherein ϕ' was encoded into an equivalent neural network. By dint of our construction, ϕ' is no longer a valid CNF (though its encoding into an MLP posed no difficulty, since any Boolean circuit is transformable into an MLP). Nevertheless, since ϕ' consists of a conjunction of but two terms, we may readily represent it as an equivalent CNF:

$$\phi' := (c_1 \vee \phi_2) \wedge (c_2 \vee \phi_2) \wedge \dots (c_m \vee \phi_2) \quad (66)$$

where each c_i is a disjunction of a few terms. Consequently, ϕ' is a valid CNF, which permits us to transform it into an equivalent random forest classifier. The reduction outlined above for MLPs applies directly to these models as well, thereby completing the proof for both model families.

□

Lemma 9. *Unless $\text{PTIME} = \text{NP}$, then there is no polynomial time algorithm for computing a subset minimal local δ -contrastive reason for either a tree ensemble or a neural network, under empirical distributions.*

Proof. We shall present a proof analogous to that of Lemma 6. Specifically, we once again avail ourselves of the classical NP-hard CNF-SAT problem defined in Lemma 6. In particular, given a Boolean formula ϕ , we demonstrate that determining a subset-minimal contrastive reason — whether for a neural network or a tree ensemble — affords us the means to decide the satisfiability of ϕ .

We first ascertain whether assigning all variables in ϕ to 1 evaluates ϕ to True. Should this be the case, the formula is satisfiable and we have determined its satisfiability. Otherwise, we invoke Lemma 8 to encode the CNF formula as a neural network f . We then compute a subset-minimal δ -sufficient reason by taking $\mathcal{D}$ to be the empirical distribution containing only a single data point $\mathbf{0}_n$, in accordance with a procedure analogous to that of Lemma 6. We additionally set $\delta = 1$ and compute a subset-minimal δ -contrastive reason with respect to $\langle f, \mathbf{x} \rangle$.

We now demonstrate that, whenever any subset-minimal δ contrastive reason is obtained for ϕ , then ϕ itself is satisfiable, whereas if no subset-minimal δ contrastive reason is obtained, then ϕ is unsatisfiable. The validity of this claim rests upon reasoning akin to that furnished in Lemma 6. Specifically, the term $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_{\bar{S}} = \mathbf{z}_{\bar{S}})$, where the distribution $\mathcal{D}$ contemplates sampling from a single datapoint, may set the probability to either 0 or 1. Furthermore, since we seek a $\delta = 1$ contrastive reason, this amounts to asking whether there exists an assignment that alters the classification of f — which corresponds to modifying the assignment of ϕ from False to True. If such an assignment exists, then ϕ is satisfiable. If, on the other hand, no subset-minimal contrastive reason exists, then no subset of features fixed to zero — with the complementary set set to ones — evaluates to true, which is tantamount to asserting that no assignment evaluates f (and consequently ϕ) to True, whence ϕ is unsatisfiable.

In order to extend the proof from neural networks to tree ensembles, we may follow the same procedure outlined in Lemma 6, encoding the CNF formula into an equivalent random forest classifier. The proof accordingly remains valid for tree ensembles, thereby bringing the argument to a close.

□

K PROOF OF THEOREM 3

Theorem 3. *Given a neural network or tree ensemble f and an empirical distribution $\mathcal{D}$ over a fixed dataset D , Algorithm 2 yields a constant $\mathcal{O}\left(\ln\left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})}\right)\right)$ -approximation, bounded by $\mathcal{O}(\ln(|D|))$, for computing a global cardinally minimal δ -contrastive reason for f , assuming feature independence. In contrast, unless $\text{PTIME}=\text{NP}$, no bounded approximation exists for computing a local cardinally minimal δ -contrastive reason for any $\langle f, \mathbf{x} \rangle$, even when $|D| = 1$.*

Proof. We partition the proof of the theorem into two lemmas, addressing, respectively, the approximation guarantee and the result concerning the absence of a bounded approximation.

Lemma 10. *Given a neural network or tree ensemble f and an empirical distribution $\mathcal{D}$ over a fixed dataset D , Algorithm 2 yields a constant $\mathcal{O}\left(\ln\left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})}\right)\right)$ -approximation, bounded by $\mathcal{O}(\ln(|D|))$, for computing a global cardinally minimal δ -contrastive reason for f , assuming feature independence.*

Proof. We shall in fact prove a stronger claim, establishing that this holds for *any* model, subject to the trivial condition that its inference time be computable in polynomial time, together with one further mild condition that we shall detail in due course — both conditions being satisfied by our neural network and tree ensemble formalizations alike.

We begin by observing that, since we are operating with empirical distributions, the computation of the global contrastive probability value function:

$$v_{\text{con}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_{\bar{S}} = \mathbf{z}_{\bar{S}})] \quad (67)$$

may be computed in polynomial time by iterating over all pairs $\mathbf{x}, \mathbf{z}$ in the dataset D , as was previously established in Proposition 2. Since Algorithm 2 performs merely a linear number of these polynomial-time queries, its total runtime is accordingly polynomial.

As regards the approximation, the classical work of Wolsey et al. (Wolsey, 1982) established a harmonic-series-based approximation guarantee for monotone non-decreasing submodular functions v with integer values — more generally, their result yields an approximation factor of $\mathcal{O}\left(\ln\left(\frac{v([n])}{\min_{i \in [n]} v(\{i\})}\right)\right)$. We showed in Proposition 2 that the value function v_{con}^g is monotone non-decreasing, while, under feature independence, Proposition 4 establishes its submodularity. Conjoined with the fact that Algorithm 2 runs in polynomial time, this directly yields an approximation guarantee of $\mathcal{O}\left(\ln\left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})}\right)\right)$.

However, we must further show that the expression $\mathcal{O}\left(\ln\left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})}\right)\right)$ is both finite and bounded by $\mathcal{O}(\ln(|D|))$, whence it is effectively constant, by dint of the assumption of a fixed dataset $|D|$. To this end, we first confirm that the expression is well-defined and finite, and thereafter proceed to establish the desired bounds. To do so, we introduce a preprocessing step in which we expunge the “redundant” elements — those liable, in theory, to render the denominator $\min_{i \in [n]} v_{\text{con}}^g(\{i\})$ equal to zero. We begin by rendering precise what is meant by redundancy:

Definition 1. *Let $\mathcal{D}$ denote some empirical distribution over a dataset D . Then we say that some feature $i \in [n]$ is redundant with respect to $\mathcal{D}$ if for any pair $\mathbf{x}, \mathbf{z} \in D$ it holds that $f(\mathbf{x}_{[n] \setminus \{i\}}; \mathbf{z}_{\{i\}}) = f(\mathbf{x})$.*

Here, the notation $f(\mathbf{x}_{[n] \setminus \{i\}}; \mathbf{z}_{\{i\}})$ signifies that all features in $[n] \setminus \{i\}$ are fixed to their values in $\mathbf{x}$, while feature i is set to its value in $\mathbf{z}$. Notably, this is tantamount to defining:

$$\forall S \subseteq [n], \mathbf{z} \in D \quad f(\mathbf{x}_S; \mathbf{z}_{\bar{S}}) = f(\mathbf{x}_{S \setminus \{i\}}; \mathbf{z}_{\bar{S} \cup \{i\}}) \quad (68)$$

As before, the notation $(\mathbf{x}_S; \mathbf{z}_{\bar{S}})$ signifies that the features in S are fixed to their values in $\mathbf{x}$, while the features in $\bar{S}$ are fixed to their values in $\mathbf{z}$. Thus, in this sense, if we “remove” feature i from the input space and define a new function f' over the reduced space $[n'] := [n] \setminus \{i\}$, then for any input of size n' the output of f' coincides exactly with the output of f when applied to the same features $[n] \setminus \{i\}$. This removal may be effected in polynomial time for both tree ensembles and neural networks: for neural networks, it amounts to detaching the corresponding input neuron from the network; for tree ensembles, to expunging any splits on that feature from all decision trees. Given the empirical nature of the contrastive value function $v_{\text{con}}^g(S)$ — as canvassed earlier in this proof and in Proposition 2 — we may compute each $v_{\text{con}}^g(\{i\})$ for all $i \in [n]$ by iterating over all pairs $\mathbf{x}, \mathbf{z} \in \mathcal{D}$ and ascertaining whether:

$$f(\mathbf{x}_{[n] \setminus \{i\}}; \mathbf{z}_{\{i\}}) \neq f(\mathbf{x}) \quad (69)$$

Should this condition hold, it attests that feature i is not redundant; conversely, should the condition fail for all pairs traversed during the iteration, then i is adjudged redundant. Once all redundant features have been identified with respect to the empirical distribution $\mathcal{D}$, we may remove them and construct an equivalent model f' — a transformation which, as discussed above, is performable in polynomial time for both neural networks and decision trees.

To bring our proof to a close, we observe that $v_{\text{con}}^g(\{i\})$ equals 0 for some empirical distribution $\mathcal{D}$ if and only if feature i is redundant with respect to $\mathcal{D}$. Consequently, once the preprocessing step has removed all redundant features from f and we have constructed the resulting function f' , we are assured that $\min_{i \in [n]} v_{\text{con}}^g(\{i\}) > 0$.

We now present the more precise bounds alluded to in our proof. Specifically, we demonstrate that the approximation factor is $\mathcal{O}(\ln(|\mathcal{D}|))$; given that the empirical distribution $\mathcal{D}$ is fixed, this yields a constant approximation. Although tighter bounds may well be attainable, our sole aim here is to establish that the bound is *constant* — a property that will stand later in sharp contrast to the local explainability setting, wherein no bounded approximation exists.

More specifically, we know that the probability function $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_{\bar{S}} = \mathbf{z}_{\bar{S}})$ is, by the very definition of empirical distributions, at least $\frac{1}{|\mathcal{D}|}$. It follows that the definition of $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_{\bar{S}} = \mathbf{z}_{\bar{S}})]$ (that is, the value function v_{con}^g) is at least $\frac{1}{|\mathcal{D}|^2}$. In particular, this further entails that:

$$\min_{i \in [n]} v_{\text{con}}^g(\{i\}) \geq \frac{1}{|\mathcal{D}|^2} \quad (70)$$

Consequently, we obtain that:

$$\ln \left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})} \right) \leq \ln(|\mathcal{D}|^2) = \mathcal{O}(\ln(|\mathcal{D}|)) \quad (71)$$

which thereby concludes our proof.

□

Lemma 11. *Unless $\text{PTIME} = \text{NP}$, there is no bounded approximation for computing a cardinally minimal local δ -contrastive reason for any $\langle f, \mathbf{x} \rangle$ where f is either a neural network or a tree ensemble, even when $|\mathcal{D}| = 1$.*

Proof. Assume, towards a contradiction, that there exists a bounded approximation algorithm for computing a cardinally minimal local δ -contrastive reason for some $\langle f, \mathbf{x} \rangle$, where f is a neural network or a tree ensemble — even when $|\mathcal{D}| = 1$. Lemma 9, however, establishes that even with a single baseline $\mathbf{z} = 0_n$ (that is, when the entire dataset comprises but one instance), adjudicating whether a contrastive reason exists is NP-hard unless $\text{PTIME} = \text{NP}$. Hence, were a polynomial-time algorithm to furnish a bounded approximation for a cardinally minimal contrastive reason, it would controvert this result, for such an approximation would implicitly decide the existence of a contrastive explanation.

□

L PROOF OF THEOREM 4

Theorem 4. *Given a neural network or tree ensemble f and an empirical distribution $\mathcal{D}$ over a fixed dataset D , Algorithm 2 yields a constant $\mathcal{O}\left(\frac{1}{1-k^f} + \ln\left(\frac{v_{\text{suff}}^g([n])}{\min_{i \in [n]} v_{\text{suff}}^g(\{i\})}\right)\right)$ -approximation for computing a global cardinally minimal δ -sufficient reason for f , assuming feature independence. In contrast, unless $\text{PTIME}=\text{NP}$, there is no bounded approximation for computing a local cardinally minimal δ -sufficient reason for any $\langle f, \mathbf{x} \rangle$, even when $|D| = 1$.*

Proof. We partition the proof into two lemmas: the one establishing the approximation guarantee for the global case, the other demonstrating the intractability of the local case.

Lemma 12. *Given a neural network or tree ensemble f and an empirical distribution $\mathcal{D}$ over a fixed dataset D , Algorithm 2 yields a constant $\mathcal{O}\left(\frac{1}{1-k^f} + \ln\left(\frac{v_{\text{suff}}^g([n])}{\min_{i \in [n]} v_{\text{suff}}^g(\{i\})}\right)\right)$ -approximation for computing a global cardinally minimal δ -sufficient reason for f , assuming feature independence.*

Proof. The proof will proceed along lines analogous to those of Lemma 10, wherein we showed that, for both neural networks and tree ensembles, Algorithm 2 attains an approximation factor of $\ln\left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})}\right)$. Once the preprocessing step has been applied, this ratio is guaranteed to be finite and bounded by $\mathcal{O}(\ln(|D|))$.

As in Lemma 10, we shall once again prove a stronger claim — namely, that the result holds for *any* model, subject to the trivial condition that its inference time be polynomially computable, together with the additional condition concerning the removal of redundant features, as described in Lemma 10. As discussed there, both conditions are satisfied by our neural network and tree ensemble formalizations alike.

Here too, since we are operating with empirical distributions, the computation of the global sufficient probability value function:

$$v_{\text{suff}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)] \quad (72)$$

may be computed in polynomial time by iterating over all pairs $\mathbf{x}, \mathbf{z}$ in the dataset D , as was previously shown in both Lemma 10 and Proposition 2. Since Algorithm 2 issues merely a linear number of such polynomial-time queries, its overall runtime is polynomial.

In contradistinction to Lemma 10, where the approximation rested upon the result of Wolsey et al. (Wolsey, 1982) for monotone non-decreasing submodular functions, the present setting demands a different condition on account of the supermodular nature of the function. Specifically, Shi et al. (Shi et al., 2021) furnished an approximation guarantee of $\mathcal{O}\left(\frac{1}{1-k^f} + \ln\left(\frac{v_{\text{suff}}^g([n])}{\min_{i \in [n]} v_{\text{suff}}^g(\{i\})}\right)\right)$, where $k^f := 1 - \min_{i \in [n]} \frac{v([n]) - v([n] \setminus \{i\})}{v(i) - v(\emptyset)}$. We established in Proposition 2 that the value function v_{suff}^g is monotone non-decreasing, while, under the assumption of feature independence, Proposition 3 further shows it to be supermodular. Given that Algorithm 2 runs in polynomial time, this issues directly in an approximation guarantee of $\mathcal{O}\left(\frac{1}{1-k^f} + \ln\left(\frac{v_{\text{suff}}^g([n])}{\min_{i \in [n]} v_{\text{suff}}^g(\{i\})}\right)\right)$, where $k^f := 1 - \min_{i \in [n]} \frac{v_{\text{suff}}^g([n]) - v_{\text{suff}}^g([n] \setminus \{i\})}{v_{\text{suff}}^g(\{i\}) - v_{\text{suff}}^g(\emptyset)}$.

To show that this expression is both bounded and constant, we pursue the same preprocessing step as in Lemma 10, wherein we remove all redundant features from f — a process which, as previously explained, is carried out in polynomial time for both neural networks and tree ensembles. As before, a feature i is strictly redundant if and only if $v(\{i\}) - v(\emptyset) = 0$. This preprocessing yields a new function f' that comports itself identically to f over the remaining features and ensures that, for every i , both $v_{\text{suff}}^g(\{i\}) > 0$ and $v_{\text{suff}}^g(\{i\}) - v_{\text{suff}}^g(\emptyset) > 0$ hold.

In order to show that this expression is both bounded and constant, we perform — in analogy with Lemma 10 — the very same preprocessing phase (which, as was explained there, is performable in polynomial time for both neural networks and tree ensembles), whereby we remove all redundant features from f . We remark that here, too, a feature i is strictly redundant if and only if $v(\{i\}) - v(\emptyset) = 0$. Hence this preprocessing phase furnishes us with a new function f' which is at once

equivalent to the behavior of f over all remaining features and such that, for any i , both $v_{\text{uff}}^g(\{i\}) > 0$ and $v_{\text{uff}}^g(\{i\}) - v_{\text{uff}}^g(\emptyset) > 0$ hold.

Now, following the same reasoning as in Lemma 10, wherein the probability term is lower bounded by $\frac{1}{|D|}$ and its expected value by $\frac{1}{|D|^2}$, we obtain that $\min_{i \in [n]} v_{\text{uff}}^g(i) \geq \frac{1}{|D|^2}$ and $v_{\text{uff}}^g(i) - v_{\text{uff}}^g(\emptyset) \geq \frac{1}{|D|^2}$. This entails, as in Lemma 10, that the first term is lower bounded by $\mathcal{O}(\ln(|D|))$, while for the second term we derive the following:

Since v_{uff}^g is supermodular (Proposition 3), we have that $v_{\text{uff}}^g([n]) - v_{\text{uff}}^g([n] \setminus \{i\})$. It follows that $1 \leq v_{\text{uff}}^g([n]) - v_{\text{uff}}^g([n] \setminus \{i\}) \leq \frac{1}{|D|^2}$, which in turn entails $1 \leq \min_{i \in [n]} \frac{v_{\text{uff}}^g([n]) - v_{\text{uff}}^g([n] \setminus \{i\})}{v_{\text{uff}}^g(\{i\}) - v_{\text{uff}}^g(\emptyset)} \leq |D|^2$ and, accordingly, $1 - |D|^2 \leq k^f \leq 0$. This yields $\frac{1}{1 - k^f} \leq |D|^2$, thereby demonstrating that the overall approximation bound predicated upon the one established by Shi et al. is constant. Although this bound may in practice be considerably smaller, our aim here is simply to establish that it remains constant — in contradistinction to the local setting, wherein no bounded approximation is achievable.

□

Lemma 13. *Unless $\text{PTIME} = \text{NP}$, there is no bounded approximation for computing a local cardinally minimal δ -sufficient reason for any $\langle f, \mathbf{x} \rangle$, even when $|D| = 1$.*

Proof. The proof proceeds along lines analogous to those of Lemma 11. Suppose, towards a contradiction, that there exists a polynomial-time algorithm furnishing a bounded approximation to a cardinally minimal local δ -sufficient reason for some $\langle f, \mathbf{x} \rangle$, where f is either a neural network or a tree ensemble — even when $|D| = 1$. Yet Lemma 6 shows that, even in the extreme case where the dataset comprises a single baseline $\mathbf{z} = 0_n$, adjudicating whether a sufficient reason exists is NP-hard unless $\text{PTIME} = \text{NP}$. Hence the existence of such an approximation algorithm would controvert this hardness result, inasmuch as it would confer the ability to decide the existence of a sufficient reason.

□

M HARDNESS OF APPROXIMATING GLOBAL CONTRASTIVE REASONS FOR NEURAL NETWORKS USING EMPIRICAL DISTRIBUTIONS

Proposition 1. *Assume a neural network $f : \mathbb{F} \rightarrow [c]$ with ReLU activations, some dataset $\mathbf{D}$, such that $\mathcal{D}$ is the empirical distribution defined over $\mathbf{D}$, and some threshold $\delta \in \mathbb{Q}$. Then unless $\text{NP} \subseteq \text{DTIME}(n^{\mathcal{O}(\log(\log(n)))})$, there does not exist a polynomial-time $\mathcal{O}(\ln(|\mathbf{D}|) - \epsilon)$ approximation algorithm for computing a cardinally minimal global contrastive reason for f concerning $\mathcal{D}$, for any fixed $\epsilon > 0$.*

Proof. The proof proceeds by way of a polynomial-time approximation-preserving reduction from a specific variant of Set Cover which, in common with the regular Set Cover, is provably hard to approximate within the factor stated in the proposition (Vazirani, 2001).

Partial Set Cover:

Input: A universe $U := \{u_1, \dots, u_n\}$, a collection of sets $S := \{S_1, \dots, S_m\}$ such that for all j $S_j \subseteq U$, and a coverage threshold $\delta \in \{0, 1, \dots, n\}$.

Output: A minimum-size subset $C \subseteq S$ such that $|\cup_{S_j \in C} S_j| \geq \delta$.

We reduce this problem to that of computing a *global*, cardinally-minimal contrastive reason S for a neural network f with respect to a distribution $\mathcal{D}$ defined over a dataset $\mathbf{D}$, together with some threshold δ' . We begin by recalling the definition of a global contrastive reason and thereafter extend it to the setting in which the explanation is evaluated over an empirical dataset $\mathbf{D}$. The definition runs as follows:

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}})] \geq \delta. \quad (73)$$

More specifically, once we assume an empirical distribution over $\mathbf{D}$, this entails that:

$$\begin{aligned}
& \mathbb{E}_{\mathbf{x} \sim \mathcal{D}} [\Pr_{\mathbf{z} \sim \mathcal{D}} (f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}})] = \\
& \frac{1}{|\mathcal{D}|} \sum_{\mathbf{x} \in \mathcal{D}} [\Pr_{\mathbf{z} \sim \mathcal{D}} (f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}})] = \\
& \frac{1}{|\mathcal{D}|} \sum_{\mathbf{x} \in \mathcal{D}} \left(\frac{\sum_{\mathbf{z} \in \mathcal{D}_{\bar{S}}(\mathbf{z}, \mathbf{x})} \mathbf{1}_{\{f(\mathbf{z}) \neq f(\mathbf{x})\}}}{\mathcal{D}_{\bar{S}}(\mathbf{z}, \mathbf{x})} \right)
\end{aligned} \tag{74}$$

where we denote $\mathcal{D}_{\bar{S}}(\mathbf{z}, \mathbf{x}) := \{\mathbf{z} \in \mathcal{D} : \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}}\}$. We now construct a ReLU neural network f . Prior to doing so, we specify the functional form that we intend the constructed neural network f to implement. For each set S_j , we introduce two vectors $\mathbf{x}^j$ and $\mathbf{y}^j$, both belonging to $\mathbb{Z}^n$ and drawn from the dataset $\mathcal{D}$, that agree on all coordinates outside S_j — that is, that share identical feature values on the complement $\bar{S}_j$. Formally:

$$\mathbf{x}_{\bar{S}_j}^j = \mathbf{y}_{\bar{S}_j}^j \tag{75}$$

and we assign strictly disparate values to the features in S_j by selecting them from two distinct vectors $\mathbf{z}$ and $\mathbf{z}'$ such that:

$$\mathbf{x}^j = (\mathbf{x}_{\bar{S}_j}^j; \mathbf{z}_{S_j}^j) \neq (\mathbf{x}_{\bar{S}_j}^j; \mathbf{z}'_{S_j}) = (\mathbf{y}_{\bar{S}_j}^j; \mathbf{z}'_{S_j}) = \mathbf{y}^j \tag{76}$$

We aim to construct f such that it satisfies the following:

$$f(\mathbf{x}^j) = f(\mathbf{x}_{\bar{S}_j}^j; \mathbf{z}_{S_j}^j) \neq f(\mathbf{x}_{\bar{S}_j}^j; \mathbf{z}'_{S_j}) = f(\mathbf{y}_{\bar{S}_j}^j; \mathbf{z}'_{S_j}) = f(\mathbf{y}^j) \tag{77}$$

Let us now suppose that both $\mathbf{x}^j$ and $\mathbf{y}^j$ occur *uniquely* in the dataset $\mathcal{D}$ across all $[n]$ coordinates — that is, that no other vector in $\mathcal{D}$ coincides with either of them on any coordinate. Under this supposition, we obtain the following relation:

$$\begin{aligned}
& \mathbb{E}_{\mathbf{x} \sim \mathcal{D}} [\Pr_{\mathbf{z} \sim \mathcal{D}} (f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{\bar{S}_j} = \mathbf{x}_{\bar{S}_j})] = \\
& \frac{1}{|\mathcal{D}|} \sum_{\mathbf{x} \in \mathcal{D}} \left(\frac{\sum_{\mathbf{z} \in \mathcal{D}_{\bar{S}_j}(\mathbf{z}, \mathbf{x})} \mathbf{1}_{\{f(\mathbf{z}) \neq f(\mathbf{x})\}}}{\mathcal{D}_{\bar{S}_j}(\mathbf{z}, \mathbf{x})} \right) = \frac{2}{2|\mathcal{D}|} = \frac{1}{|\mathcal{D}|}
\end{aligned} \tag{78}$$

Here, the factor of 2 appearing in both the numerator and the denominator arises from the circumstance that only two vector pairs — $(\mathbf{x}^j, \mathbf{z}^j)$ and $(\mathbf{z}^j, \mathbf{x}^j)$ — contribute to the respective sums.

We reiterate this construction for each S_i in the set-cover instance, on each occasion selecting two *distinct* vectors $\mathbf{x}^i$ and $\mathbf{y}^i$ that differ on *every* coordinate in $[n]$. In sum, this yields $2m$ vectors that constitute the dataset $\mathcal{D}$ from which we construct f .

We then construct a neural network f that simulates this function. This is accomplished by recourse to a known result on *neural network memorization* (Vardi et al., 2022), which guarantees that a ReLU network capable of representing any finite set of input–output pairs may be constructed in polynomial time. Availing ourselves of this result, we construct f accordingly and set $\delta' := \frac{\delta}{|\mathcal{D}|}$.

We now contend that a subset S_j is a global contrastive reason for f under tolerance δ' if and only if C is a partial set cover for U under tolerance δ . By Equation 78, each constructed pair of vectors $(\mathbf{x}^j, \mathbf{y}^j)$ contributes exactly $\frac{1}{|\mathcal{D}|}$ to the value term. Moreover, since each S_j employs its own distinct pair of vectors in $\mathbb{Z}^n$, there is no overlap between these pairs; consequently, the coverage attained by any set cover C corresponds exactly to the accumulated contribution of the selected subsets in the input domain of f . This completes the reduction.

□

N TRACTABILITY RESULTS FOR ORTHOGONAL DNFS

Lemma 14. *If f is an orthogonal DNFS and the probability term v_{suff}^g can be computed in polynomial time given the distribution $\mathcal{D}$ (which holds for independent distributions, among others), then obtaining a subset-minimal global δ -sufficient reason can be obtained in polynomial time.*

Proof. We begin by rendering precise the notion of orthogonal DNFs (Crama & Hammer, 2011) before presenting the full proof.

Definition 2 (Orthogonal DNFs). *A Boolean function φ is in orthogonal disjunctive normal form (orthogonal DNF) (Crama & Hammer, 2011) if it is a disjunction of terms $T_1 \vee T_2 \vee \dots \vee T_m$, such that for every pair of distinct terms T_i and T_j ($i \neq j$):*

$$T_i \wedge T_j \models \perp$$

This signifies that no single variable assignment can satisfy more than one term simultaneously.

The proof mirrors the argument deployed in Theorem 1, resting upon the fact that v_{suff}^g is computable in polynomial time for any set S .

For a given set S , let $\mathcal{C}^S T_i$ denote the constraints that the term T_i imposes upon the variables in S . We enumerate all pairs of terms (T_i, T_j) ; any pair whose constraints on S are *inconsistent* (i.e., $\mathcal{C}_{T_i}^S \wedge \mathcal{C}_{T_j}^S \models \perp$) is disregarded, since it contributes nothing to v_{suff}^g .

For each *consistent* pair of terms (T_i, T_j) , we evaluate $\Pr(\mathbf{x}_S \in \mathcal{C}_{T_i, T_j}^S)$, where $\mathcal{C}_{T_i, T_j}^S = \mathcal{C}_{T_i}^S \cap \mathcal{C}_{T_j}^S$; the contribution of this pair to v_{suff}^g is then given by

$$\Pr(\mathbf{x}_S \in \mathcal{C}_{T_i, T_j}^S) \cdot \Pr(\mathbf{x}_{\bar{S}} \in \mathcal{C}_{T_i}^{\bar{S}}) \cdot \Pr(\mathbf{x}_{\bar{S}} \in \mathcal{C}_{T_j}^{\bar{S}}).$$

Under feature independence, these probabilities factorize as $\Pr(\mathbf{x}_S) = \prod_{i \in S} p(x_i)$. Since there are at most m^2 such term pairs, the value of v_{suff}^g is computable in polynomial time under an independent distribution $\mathcal{D}$. $\square$

O DISCLOSURE: USAGE OF LLMs

An LLM was employed exclusively in the capacity of a writing assistant, tasked with refining grammar and typographical errors and with enhancing clarity. It contributed in no way to the generation of research ideas, the study design, the data analysis, or the interpretation of results, all of which were carried out solely by the authors.